CSE190/291
How The Web Tracks You / Targeted Advertising

Stateful Tracking, Part 2

UC San Diego

Housekeeping

General course things to know

- PA1 part 1 grades released, yay!
- PA1, parts 1 + 2 are released!
  - Part 2 due Tuesday, 10/13 (one week from today!)
- Implementing a full-scale “web application” that does some stateful tracking, using cookies, to track your behavior on a Google search competitor (Boogle)
  - Cookie-based tracking, databases, and server side logic
- **No class on 10/13!**
  - work on your PAs :)
- PA1, parts 3 + 4 *should* release on **10/14**

Previously on CSE190/CSE291…

- We talked about the details of web pages, servers, clients, and the construction of web pages (resources, origins, first-party / third-party, oh my)
- We began our discussion about cookies, how they work, and how they’re scoped



Let’s do a recap.

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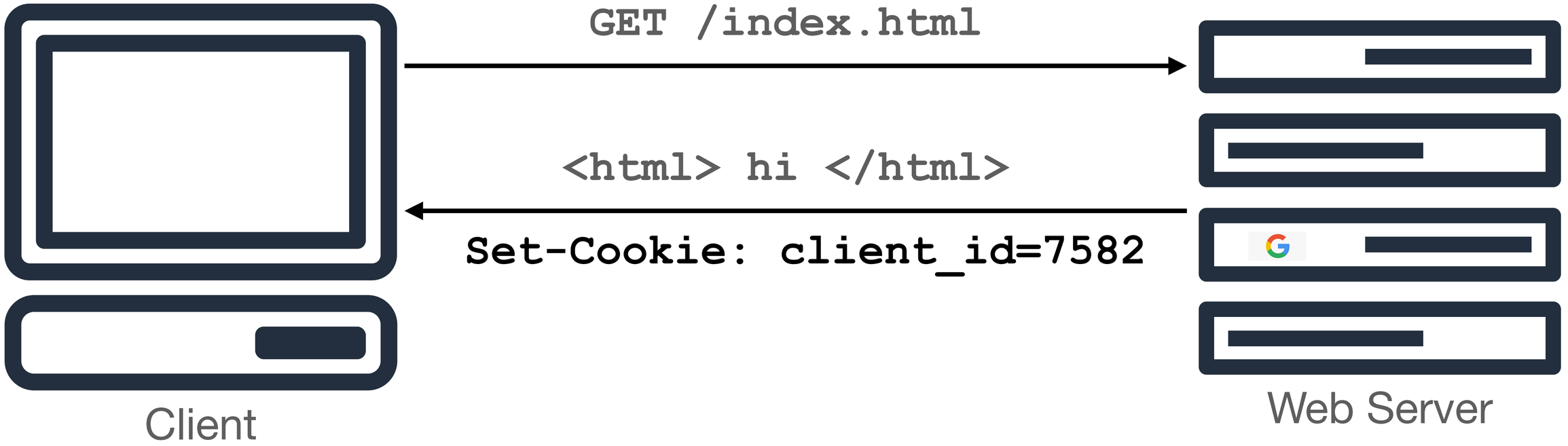
How do web cookies work?



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Client

The web client (browser) stores the cookie value in a local file.

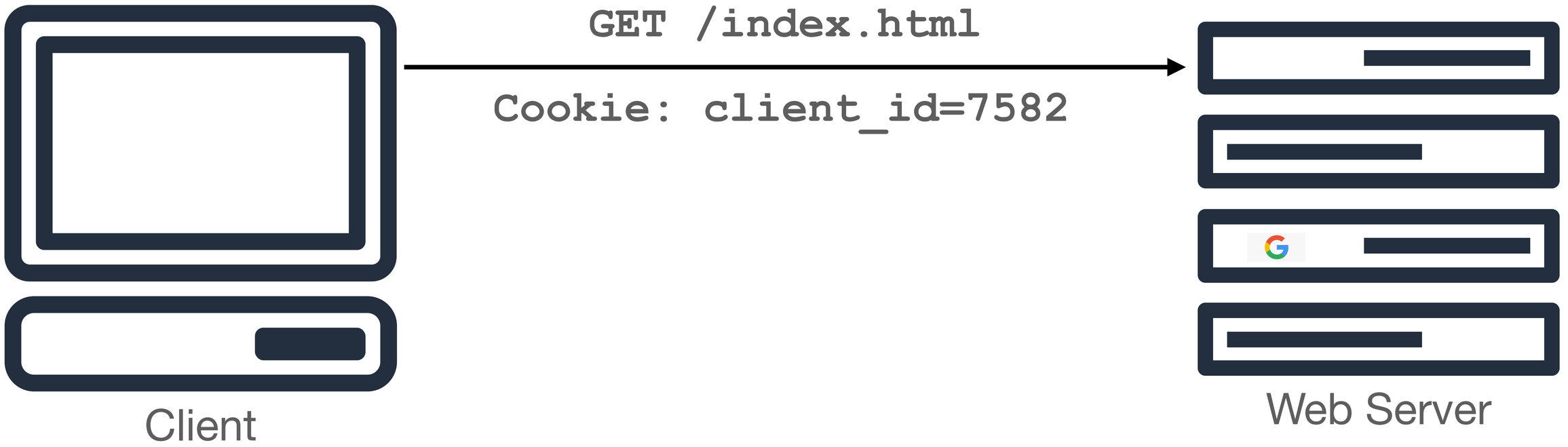


Web Server

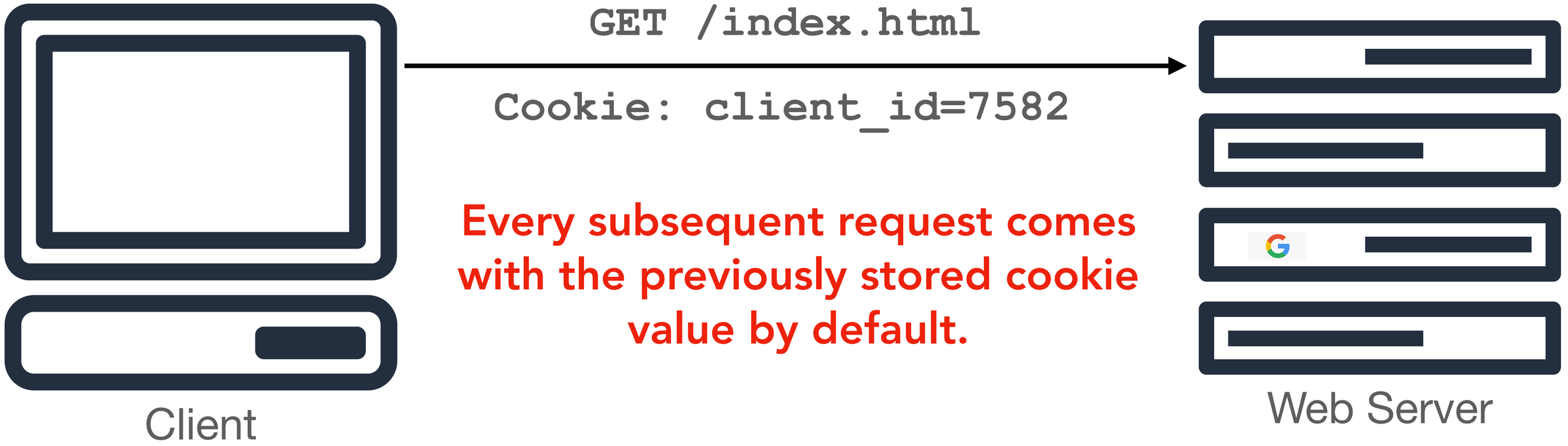
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Every subsequent request comes with the previously stored cookie value by default.

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- Does the browser automatically send cookies to *any* server without checking anything?
  - No! Cookies have scopes!

Cookie Scoping

- How cookies are shared are ultimately **scoped** by the server, by several properties:
  - Domain — `Set-Cookie: client_id=7582; Domain=example.com`
  - Path — `Set-Cookie: client_id=7582; Path=/docs`
  - Secure — `Set-Cookie: client_id=7582; Secure`
  - How they're accessed — `Set-Cookie: client_id=7582; HttpOnly`
  - Time — `Set-Cookie: client_id=7582; Max-Age=3600`

Lots of other properties…

- `SameSite` (primarily a defense against a CSRF attack — see CSE 127)
- `Partitioned` (we'll talk about this in depth later)
- Thought experiment: **when a server gets a request with a cookie, what does it see?**

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- Cookie prefixes to the rescue!
  - E.g., Prepending your cookie name with `__Host-` tells the browser to do something special, and the server knows its original intention

How do we decide to send cookies? (Exam style)

Cookie 1:
name = mycookie
value = value
domain = login.site.com
path = /

Cookie 2:
name = mycookie2
value = value
domain = site.com
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Cookie 3:
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Today's lecture — Stateful Tracking, Pt. 2

Learning Objectives

- Begin to understand the stateful, third-party tracking ecosystem, the underlying cookies that support it, and what we might do about it
- Understand the fight and history behind third-party tracking and the fights against the fights against third-party tracking
- Learn about how, despite our best efforts to block cookies, cookies remain passed around the web

Stateful Tracking

Stateful Tracking Basics

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    - Where you visit, etc.
- **Goal of most trackers: To learn about your web history (where you’ve been) and use that to build a profile about you over time.**

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Defining First-Party Tracking

- *"The site that you intended to visit tracks things about you."*
- Is this a privacy violation?
  - Not always — when you visit a website, you are "consenting" (in some implicit way) for the website to do whatever it wants with your interactions
  - Most browsers don't do anything about first-party tracking (e.g., cookies), because, well, that's just the Internet
- First-party tracking can still get companies into trouble though…

First-Party Problems



TECH

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

By Kashmir Hill, Former Staff. Welcome to The Not-So Private Parts where technology & privacy collide

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Companies might sell your data

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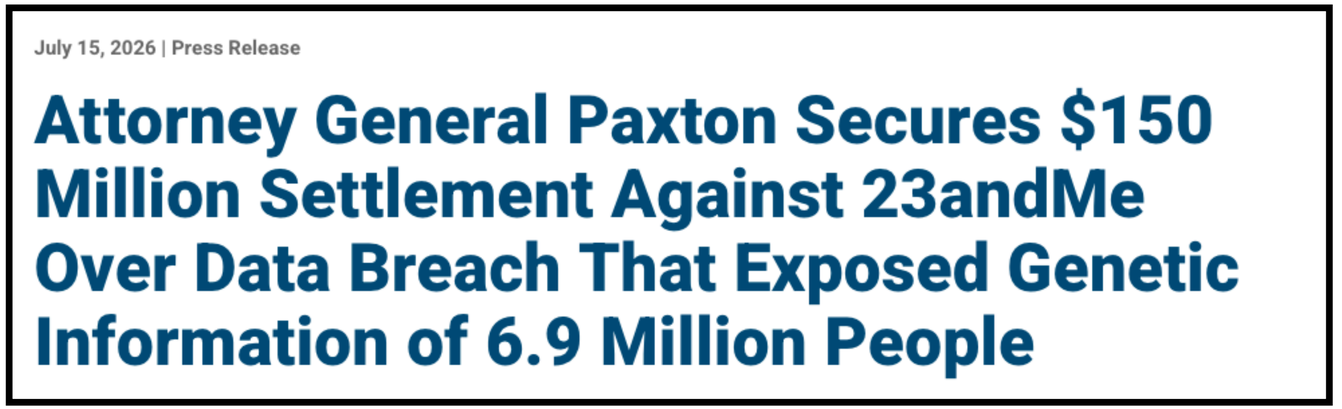


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Companies might sell your data



July 15, 2026 | Press Release

Attorney General Paxton Secures $150 Million Settlement Against 23andMe Over Data Breach That Exposed Genetic Information of 6.9 Million People

Companies might get breached
(and then go bankrupt)

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  - **What is third-party tracking?**

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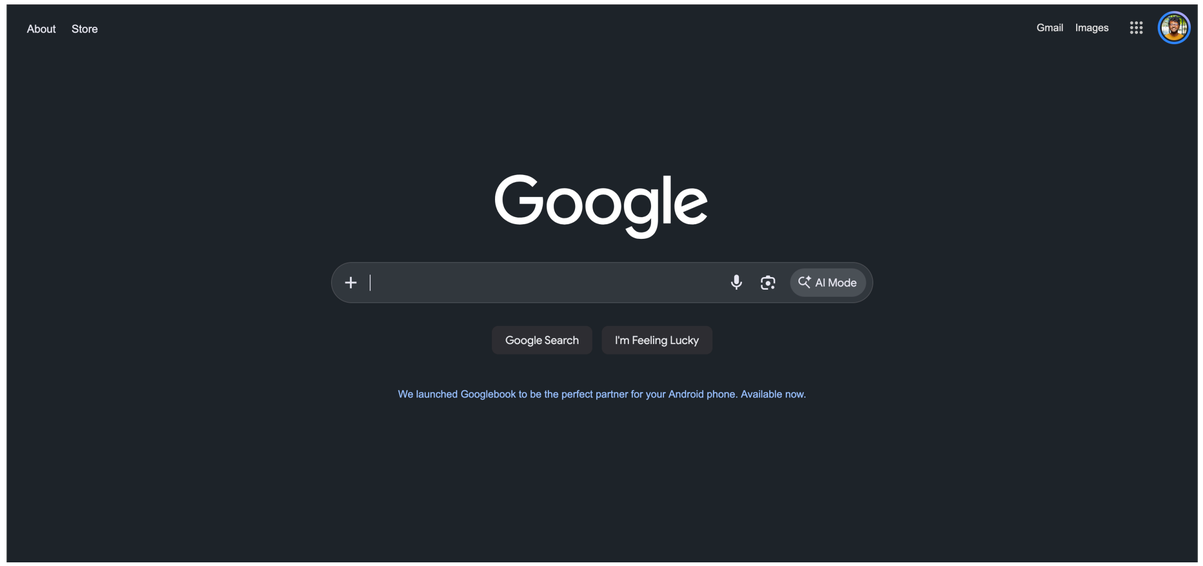
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GET google.com/



google.com

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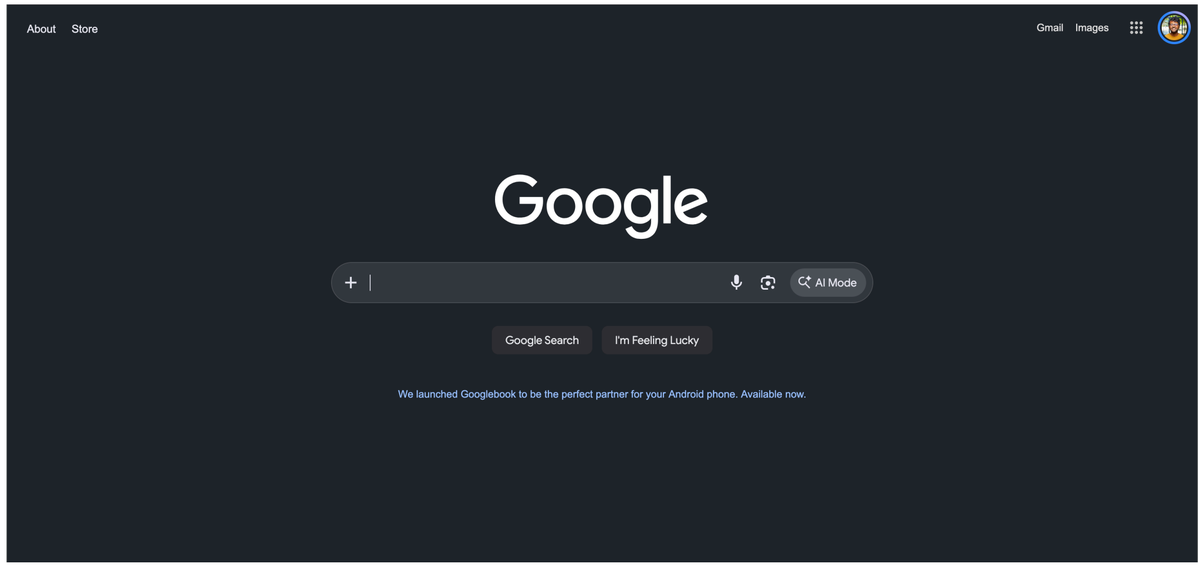


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GET google.com/

Here's the page..

`Cookie: client_id=7582`



google.com

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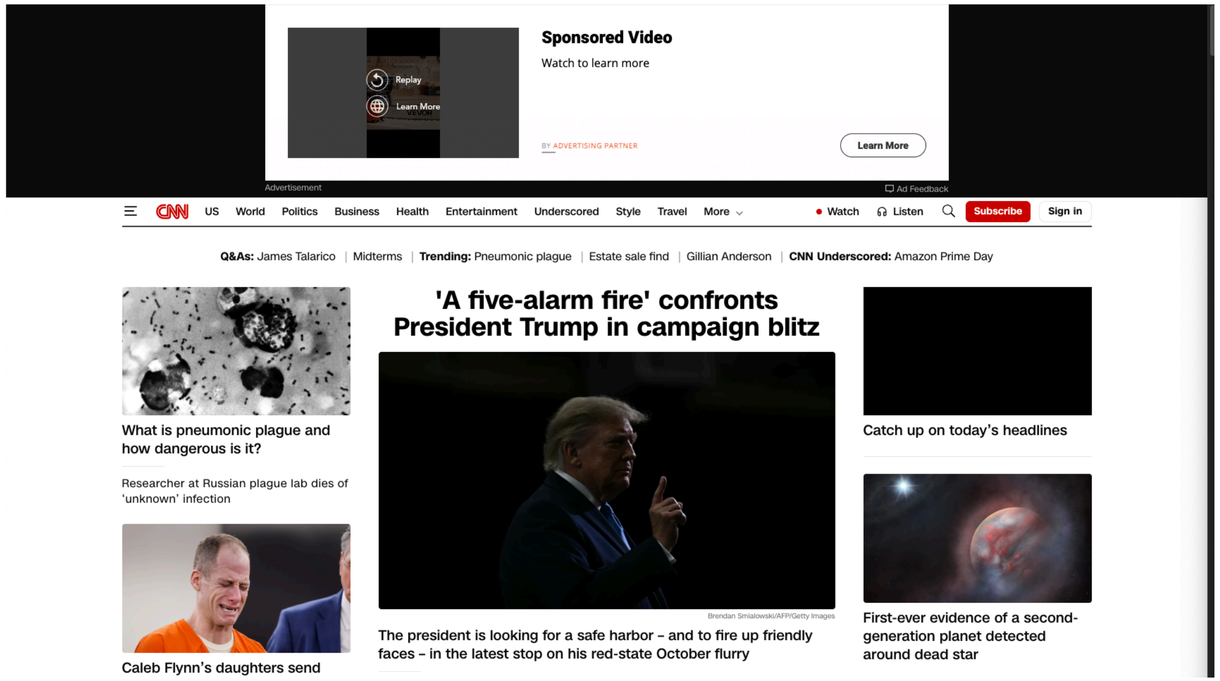
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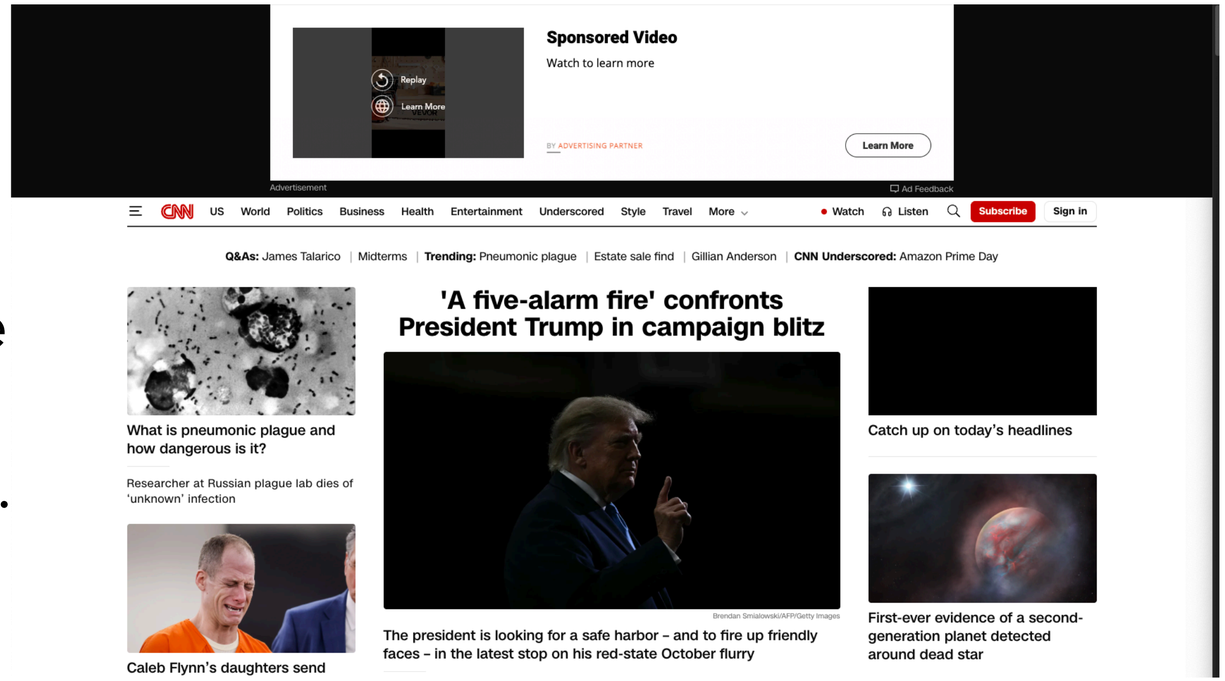
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By the time this page has loaded, I have already been tracked. How?



cnn.com

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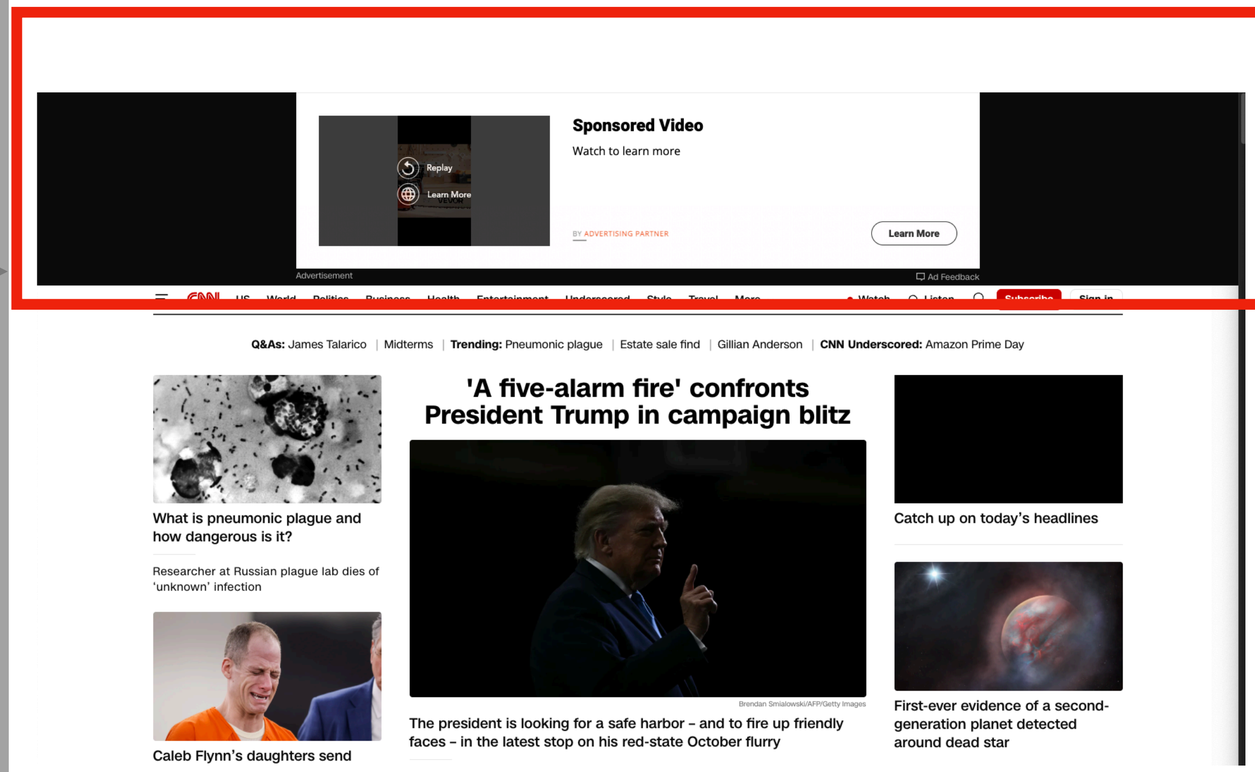
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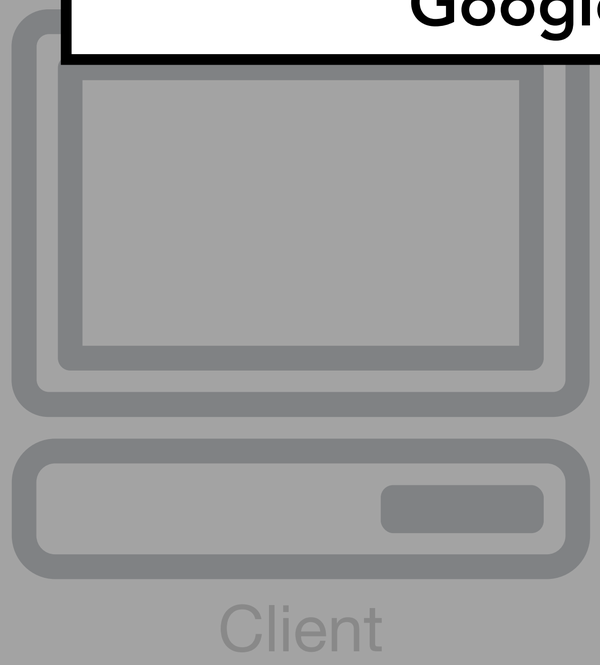
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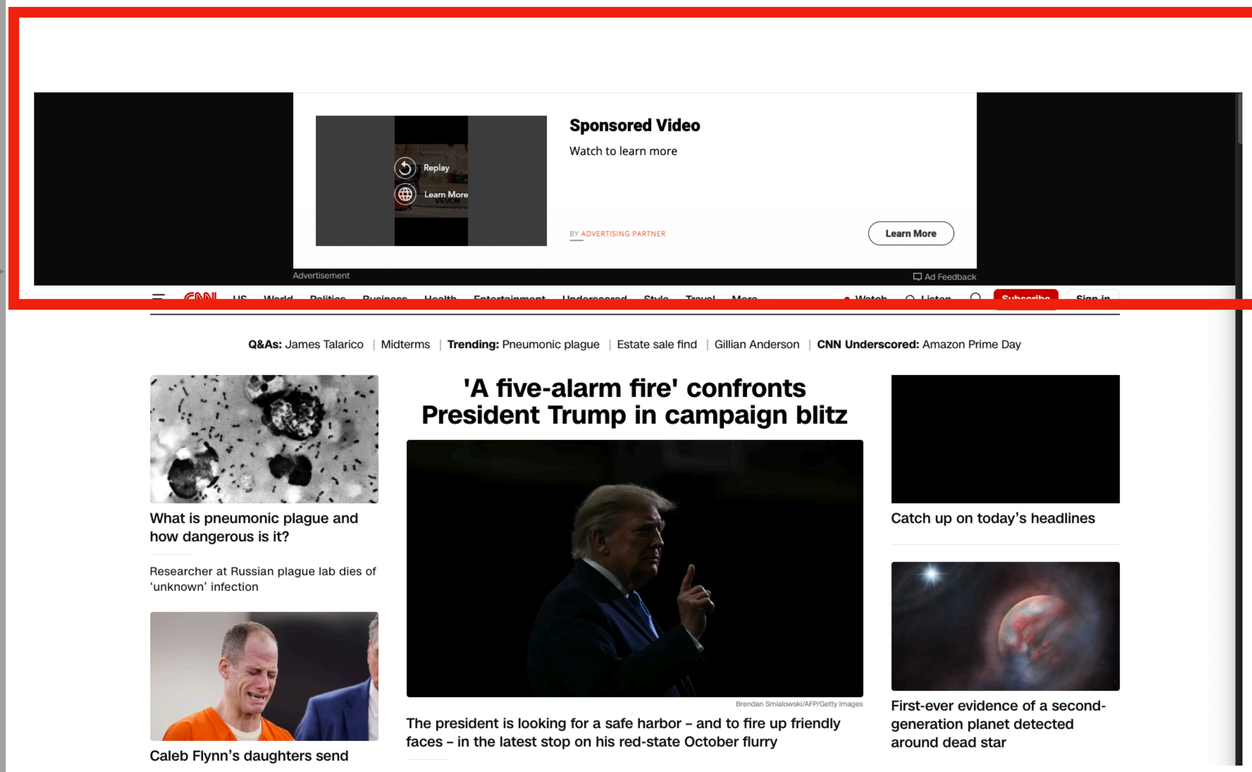
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Where did this ad load from?

Google.





cnn.com

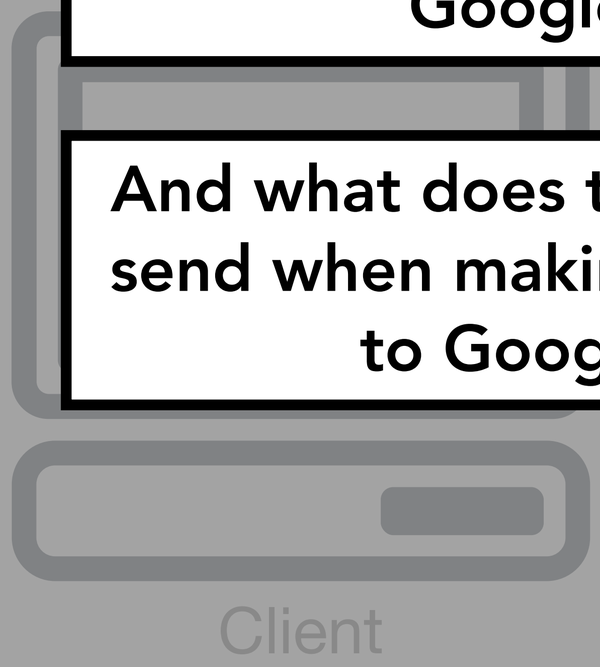
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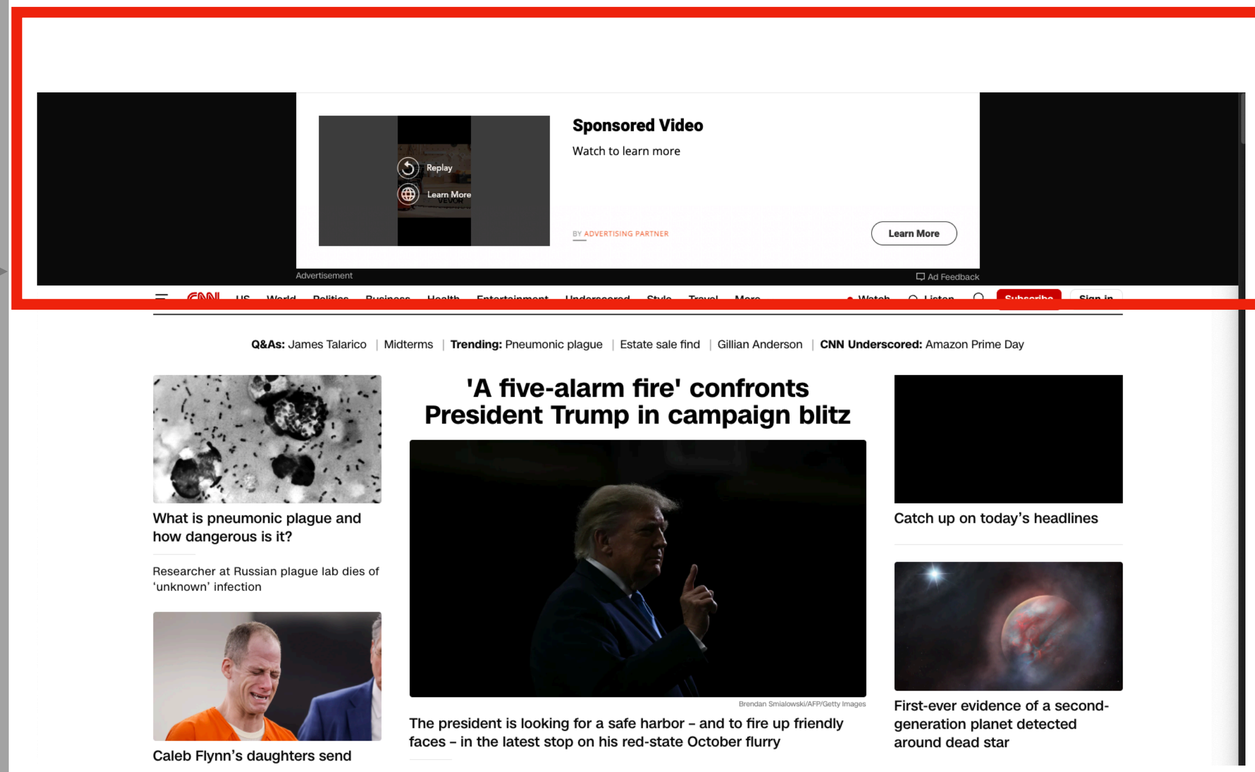
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Where did this ad load from?

Google.

And what does the browser send when making requests to Google?





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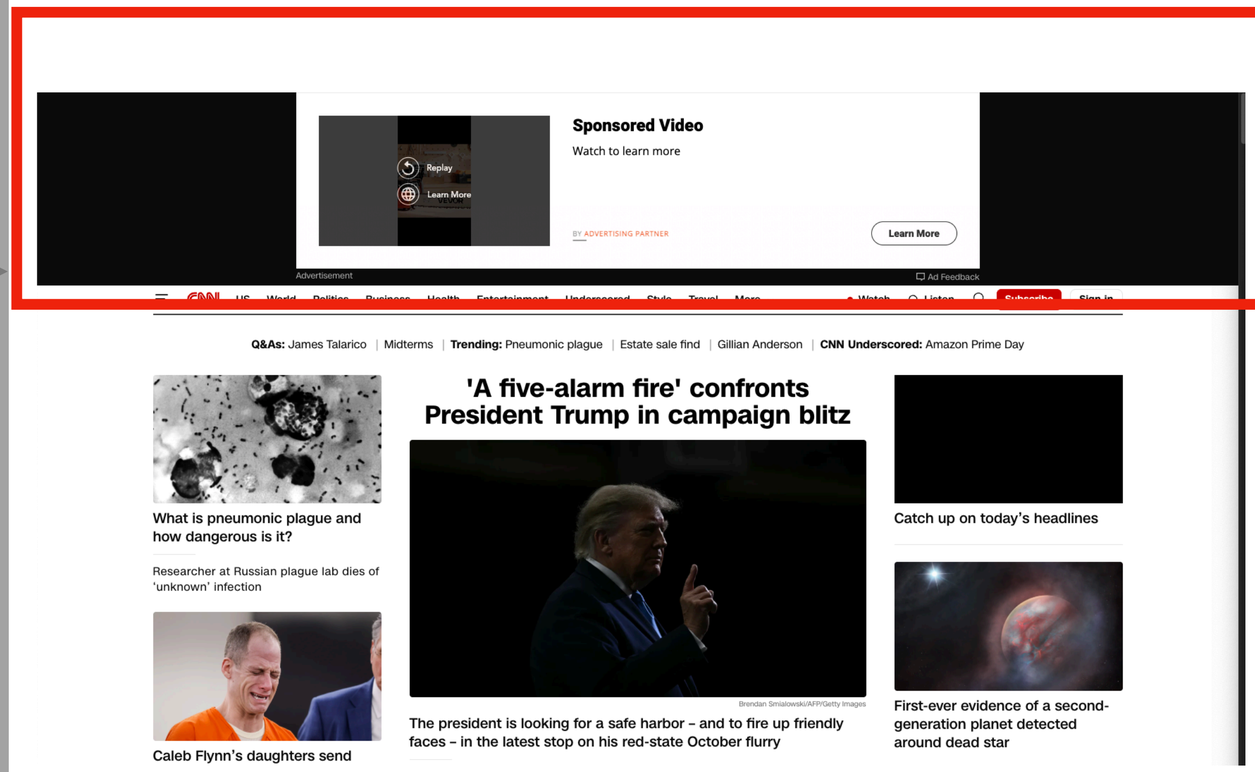
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Client



cnn.com

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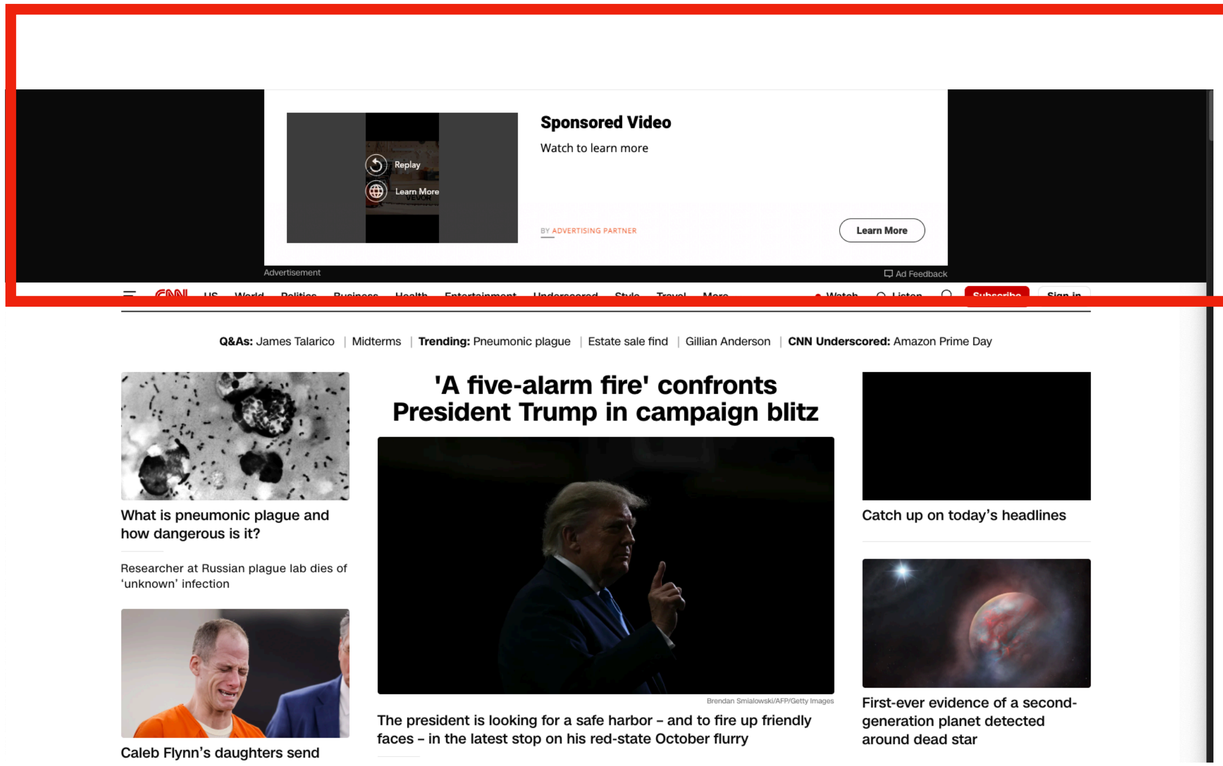
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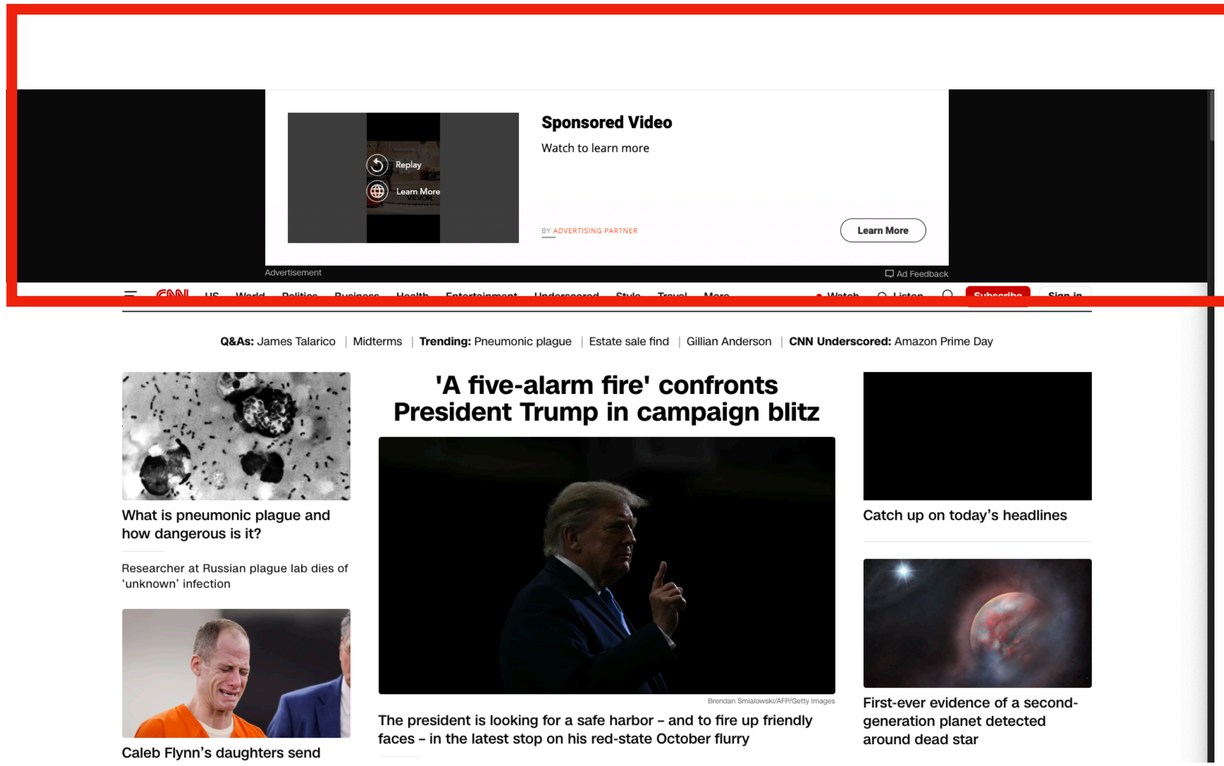


Client

GET google.com/ad

`Cookie: client_id=7582`

`Referer: cnn.com`



cnn.com

What does Google now know about me?

What does Google now know about me?

- They know I went to Google (and all my searches)
- They know I went to cnn.com
  - If I *click* on anything on cnn.com, they can know that too (assuming there's an ad on cnn.com)
- They know *when* I go anywhere on the Internet, assuming they have *any* kind of digital presence (e.g., an ad *requested* on the page)

Surely, Google can’t be in that many places, right?

Surely, Google can't be in that many places, right?

Wrong

- Major companies have large presences on the web, and as a result, can see the majority of websites that you visit
  - Google appears on **82.2%** of the Top 1M, primarily because of analytics and advertising services
  - Facebook appears on **34.1%**, to enable social sharing + tracking

Company	Prevalence on Top 1M
Google	82.2%
Facebook	34.1%
Amazon	32.6%
Cloudflare	30.7%
Akamai	20.3%
MaxCDN	19.0%
Edgecast	17.9%
Fastly	15.5%
SoftLayer	11.8%
Twitter	11.2%

Third Party Tracking is Everywhere

- Third-party cookies — cookies sent during a browsing instance that are *not to the page you're visiting*
- Not just ads! How else can this type of tracking be enabled?
  - Images, invisible pixels, redirects… basically **anything that makes the browser make an HTTP request**

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- Not just ads! How else can this type of tracking be enabled?
  - Images, invisible pixels, redirects… basically **anything that makes the browser make an HTTP request**
- Third-party tracking has been, for most of the history Internet, the de-facto way to track and follow you around as you browse
  - It happens transparently (without your knowledge) and is **built into the web ecosystem by default**

Maybe we're not doomed?

- With a group, discuss the following question:
  - In what ways is this type of stateful, third-party tracking *limited* in its ability to track you?

Limitations of vanilla third-party tracking

- Each tracker only sees its own slice of the web
  - E.g., Facebook's tracking is "less comprehensive" than Google's tracking, just based on their vantage point
- Users might install extensions, or other software that block third-party requests altogether (some browsers do this, for example!)
- Users might periodically delete their cookies, forcing new `Set-Cookie` headers and therefore breaking linkages

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All of these have been cleverly bypassed by the ecosystem over time.

Bypassing Cookie Protections

Break Time + Attendance



Codeword:
ChipsAhoy

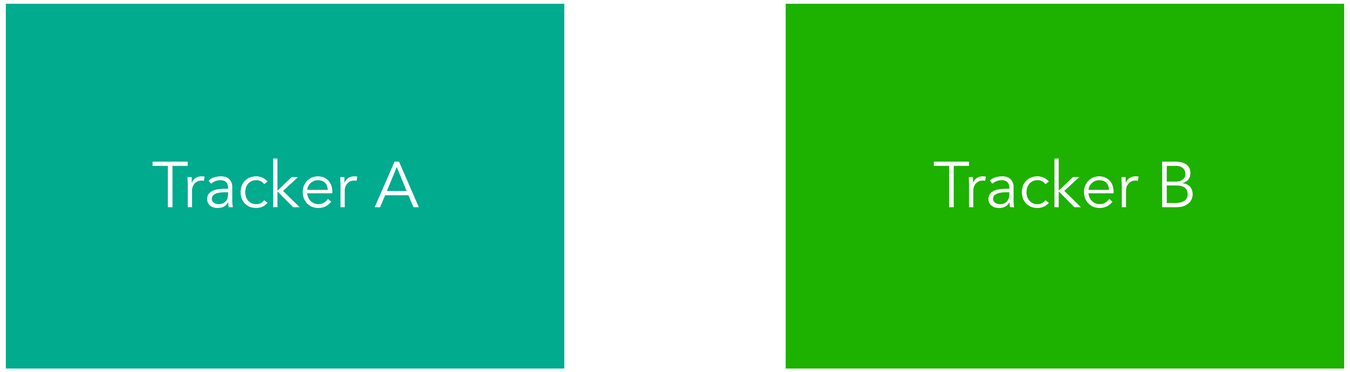
https://tinyurl.com/webtracking-attendance

A tale of two trackers

- Problem — TrackerA and TrackerB only see their own slice of the world, but they'd like to see more

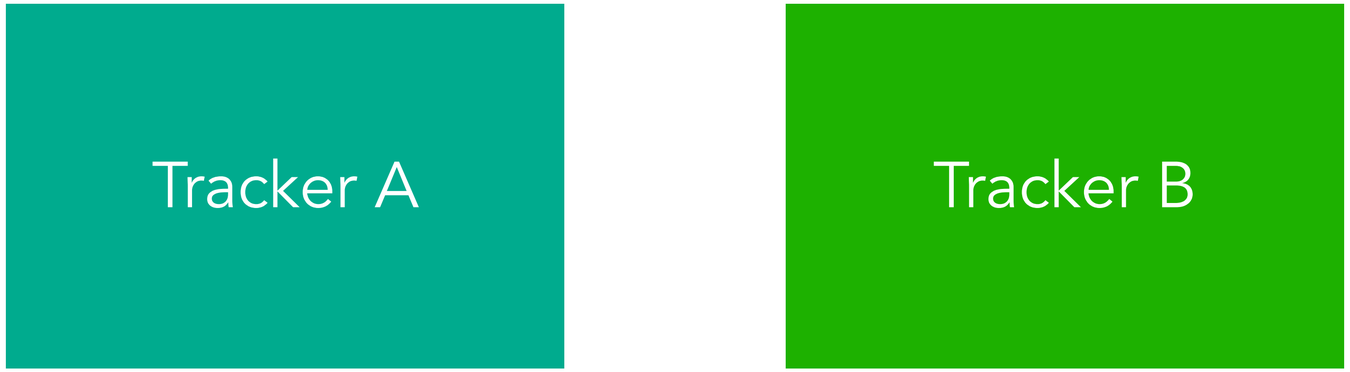
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- NYTimes
- CNN
- HuffingtonPost

- HackerNews
- TechCrunch
- Wired

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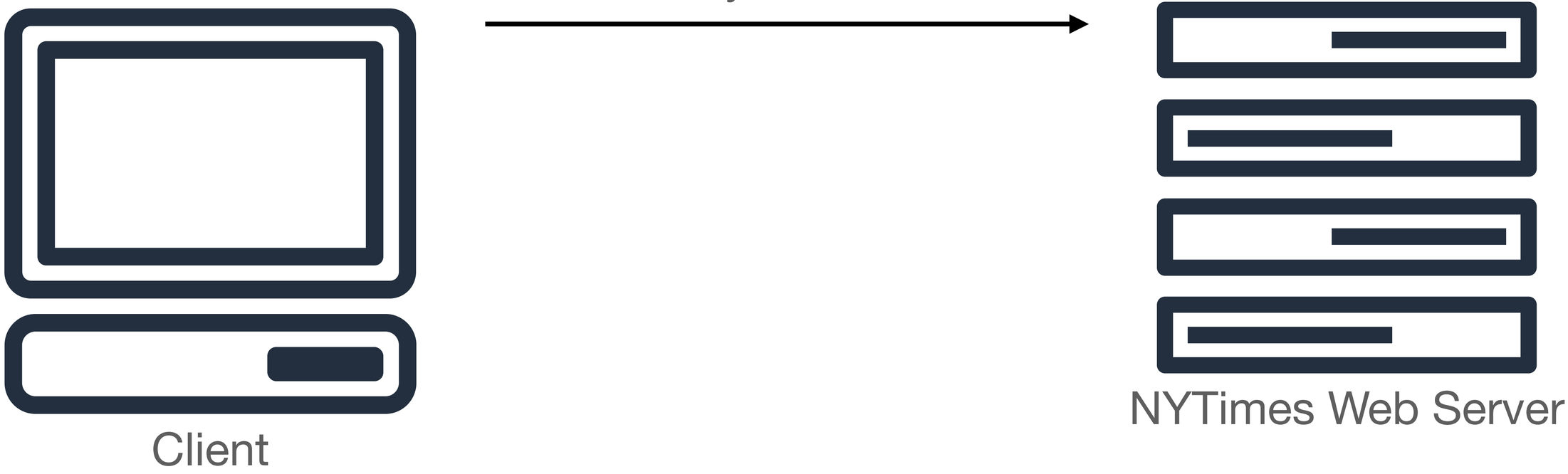
Tracker A
t_id=ABC

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Tracker B
t_id=123

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TrackerA



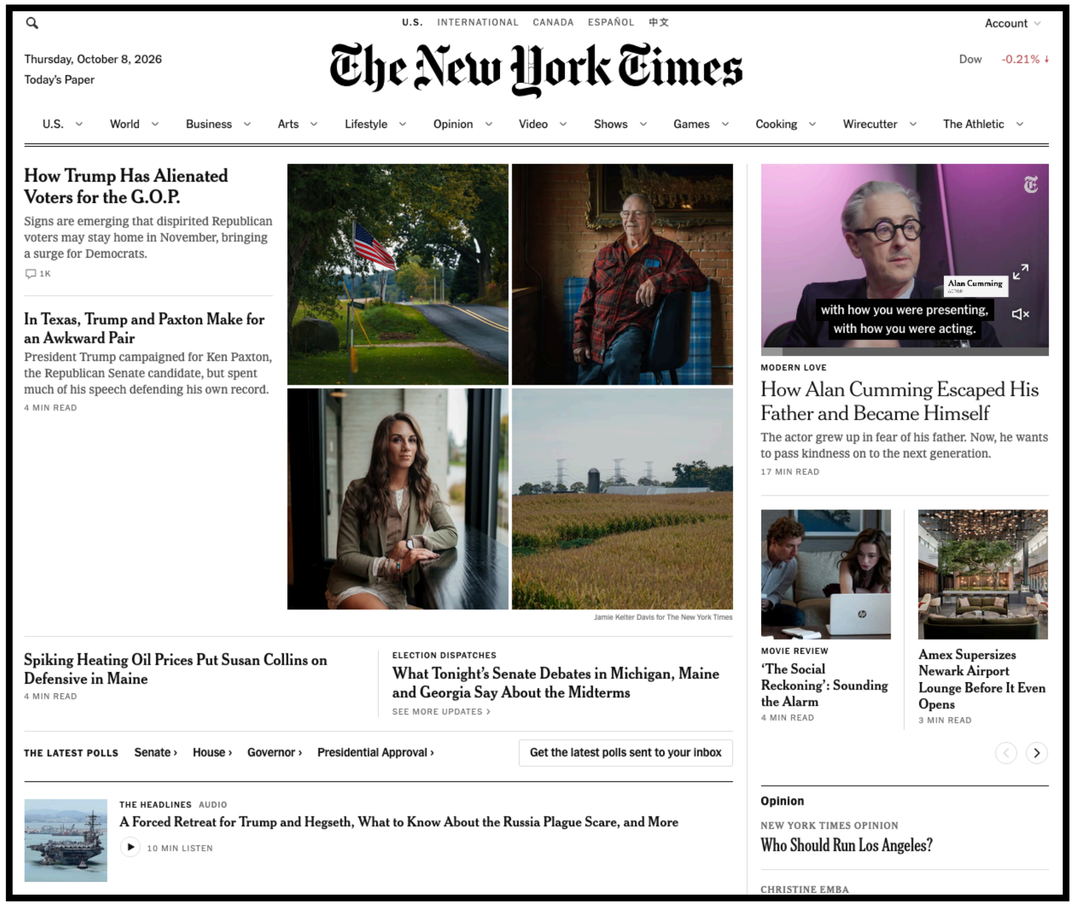
TrackerA



Client

GET nytimes.com/

NYTimes Response



TrackerA

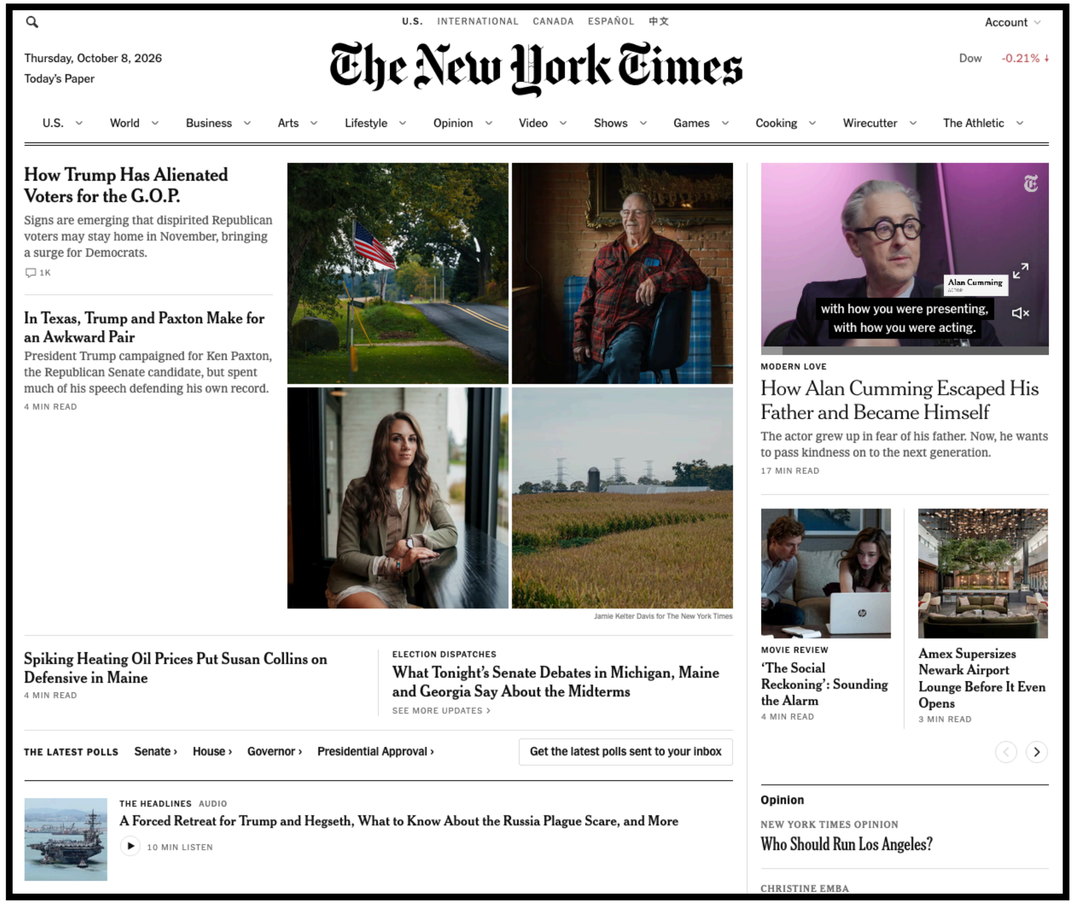


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TrackerA



Client

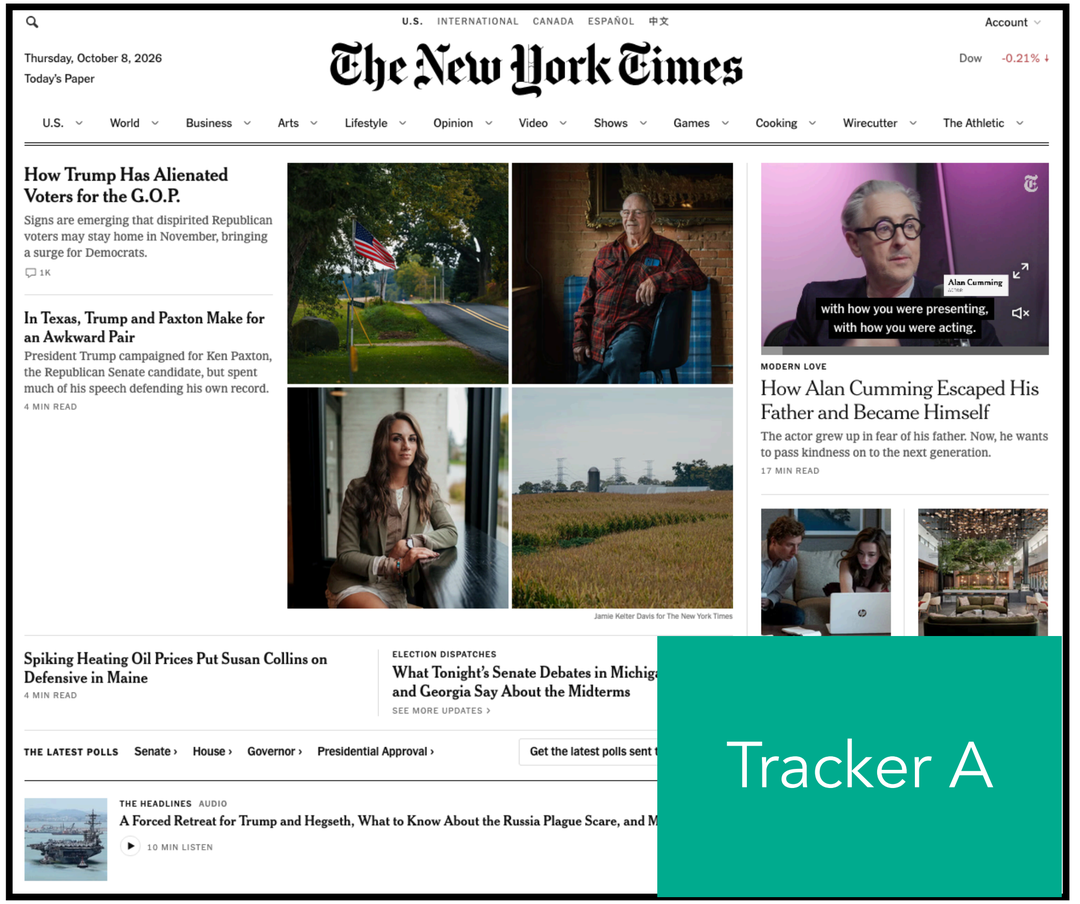
GET nytimes.com/

NYTimes Response

GET TrackerA

TrackerA Pixel

`Set-Cookie: t_id=ABC`



TrackerB



Client

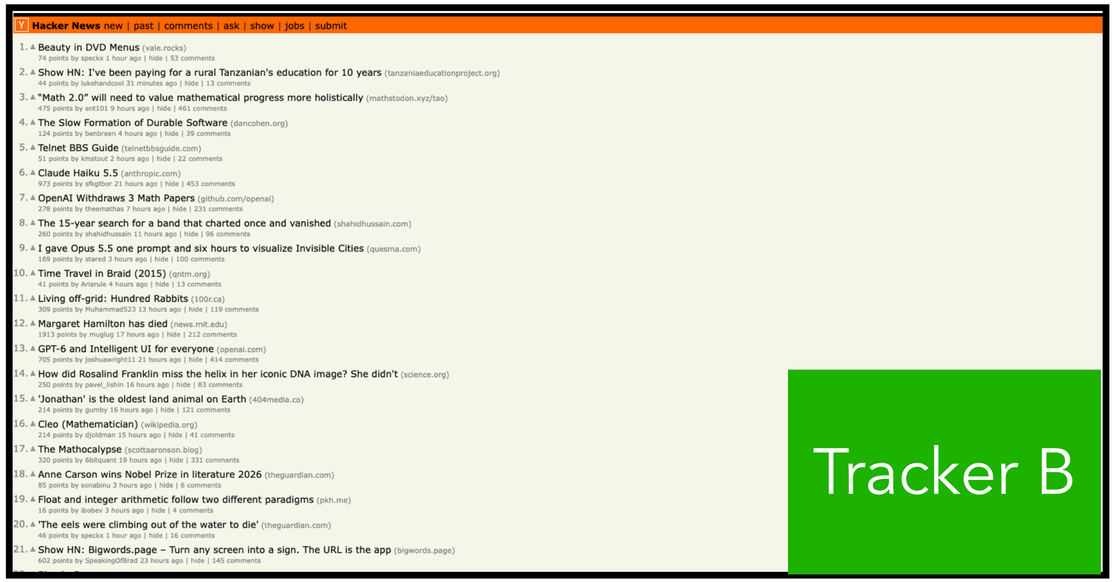
GET HackerNews

HN Response

GET TrackerB

TrackerB Pixel

`Set-Cookie: t_id=123`



TrackerA 🤝 TrackerB

- TrackerA and TrackerB have different identifiers for the same user!
  - So how can TrackerA and TrackerB work together to use their vantage points to build a complete picture of each user?
- They can independently agree to **synchronize** cookies automatically when the browser makes requests — through a tactic called *Cookie Syncing*

TrackerA 🤝 TrackerB



`GET TrackerA`

Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client

```
GET TrackerA
```

```
302 trackerb.com/sync?t_id=ABC
```

Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client

```
GET TrackerA
```

```
302 trackerb.com/sync?t_id=ABC
```

```
GET trackerb.com/sync?t_id=ABC
Referer: TrackerA
```

Tracker A

Tracker B

TrackerA 🤝 TrackerB



Client

```
GET TrackerA
```

```
302 trackerb.com/sync?t_id=ABC
```

```
GET trackerb.com/sync?t_id=ABC
Referer: TrackerA
```

Tracker A

Tracker B

TrackerB now knows TrackerA's cookie identifier, and can match in a DB on their backend!

What just happened?

- TrackerA and TrackerB had disparate **cookies** for the same browser
- But after cookie syncing, **both** now have a linkage between what “TrackerA” calls this person and what “TrackerB” calls this person
- Question: Is cookie syncing restricted to just two trackers?

What just happened?

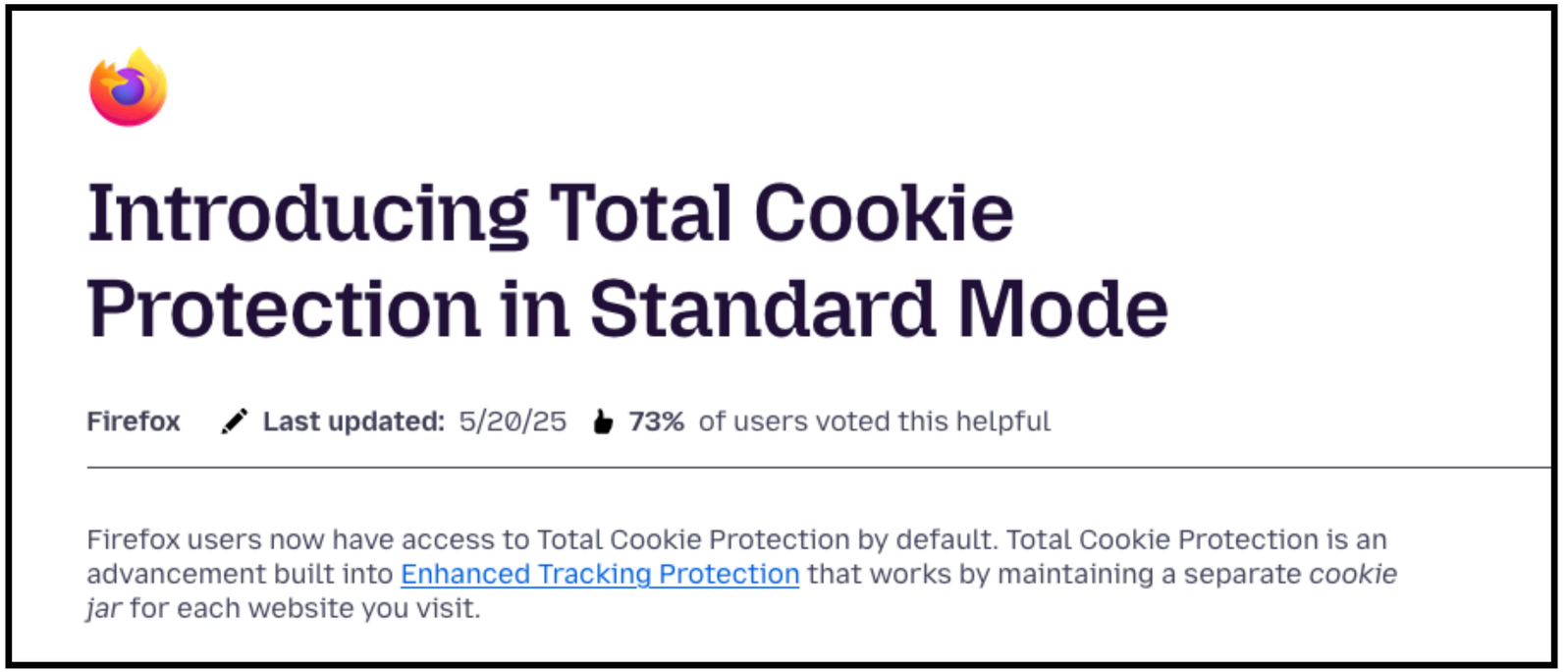
- TrackerA and TrackerB had disparate **cookies** for the same browser
- But after cookie syncing, **both** now have a linkage between what “TrackerA” calls this person and what “TrackerB” calls this person
- Question: Is cookie syncing restricted to just two trackers? **No!**
- Why does this matter?
  - It defeats browser protections put in place by cookie scoping
  - Your data is traveling further than you might expect
  - It relies on underlying browser feature that simply *has* to work for the web to function (redirects)

Cookie Syncing in the Wild

- See Demo

Defeating Cookie Syncing – Partitions

- Many browsers (Safari, Firefox, Brave) have gotten fed up with all this tracking, and implemented **cookie partitioning**
- For example — Firefox calls this Total Cookie Protection (other browsers call it other things), released in **2021**



Introducing Total Cookie Protection in Standard Mode

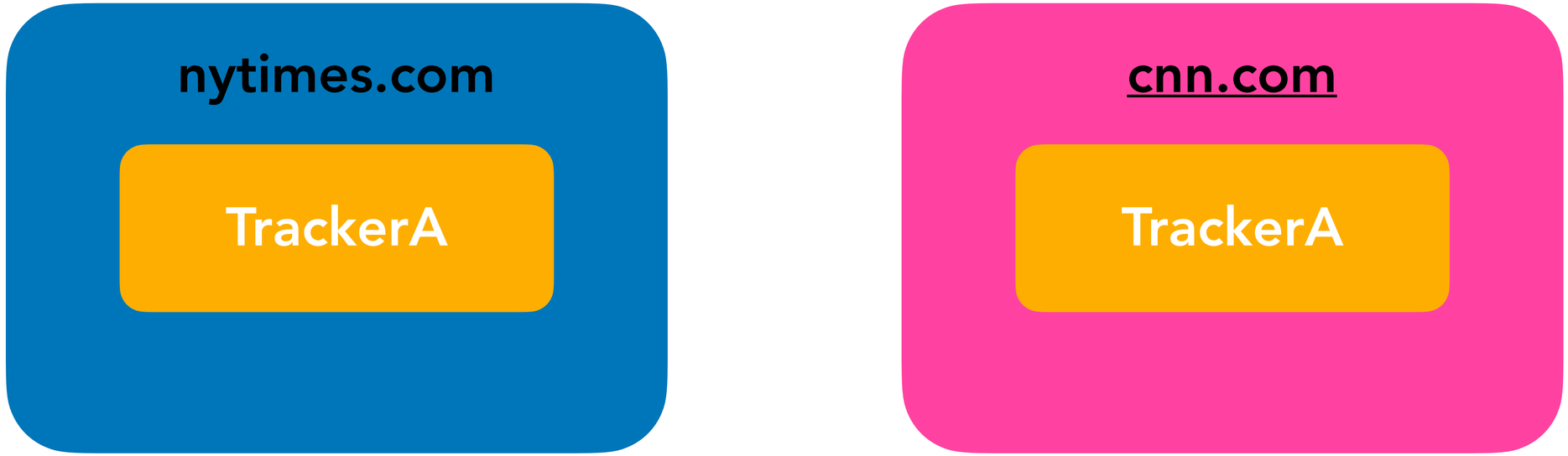
Firefox **Last updated:** 5/20/25 **73%** of users voted this helpful

Firefox users now have access to Total Cookie Protection by default. Total Cookie Protection is an advancement built into Enhanced Tracking Protection that works by maintaining a separate *cookie jar* for each website you visit.

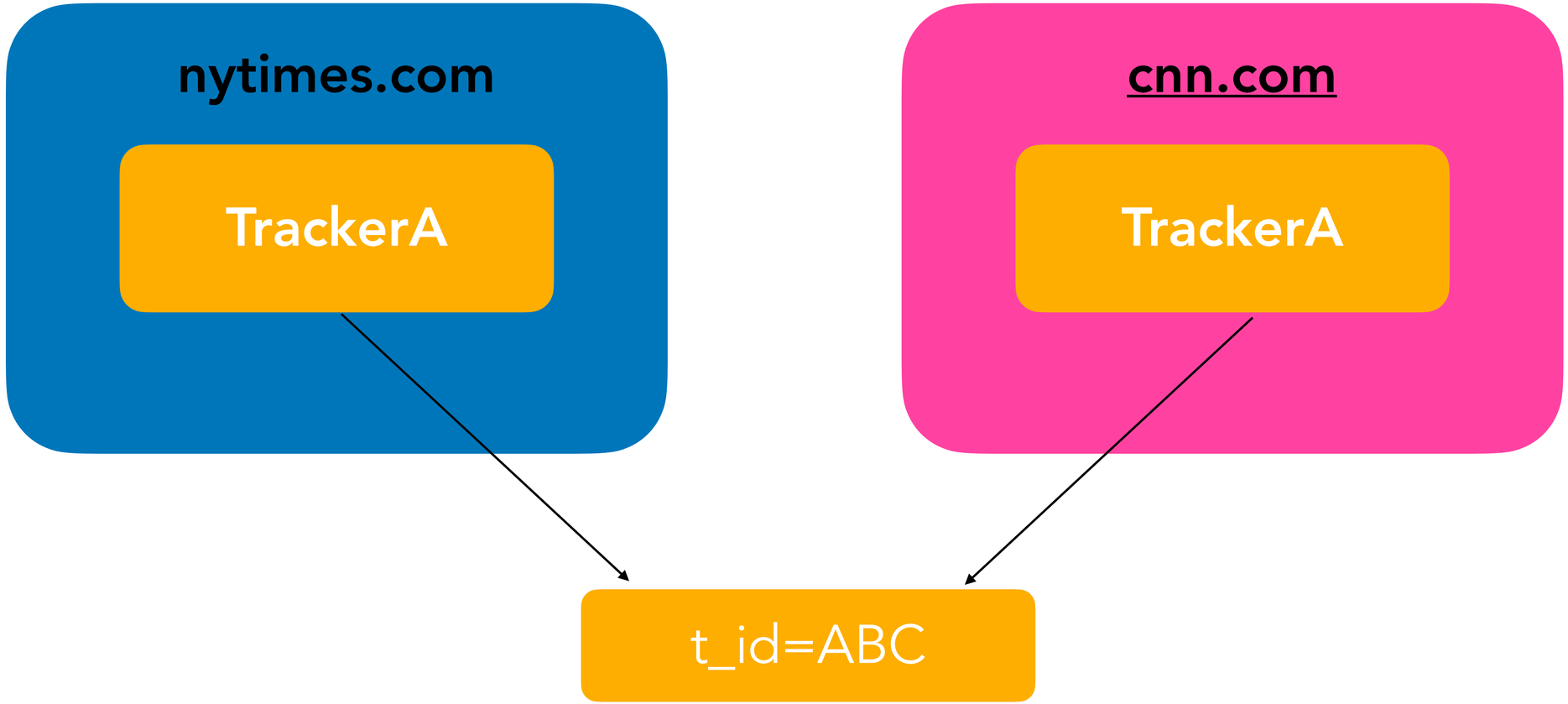
Cookie Partitions

- Instead of having cookie scope *solely* determined by “which domain sets the cookie” — have it also scoped by **the top-level page**
  - In order for a browser to send a cookie, it must match **both** the top-level page *and* the third-party domain

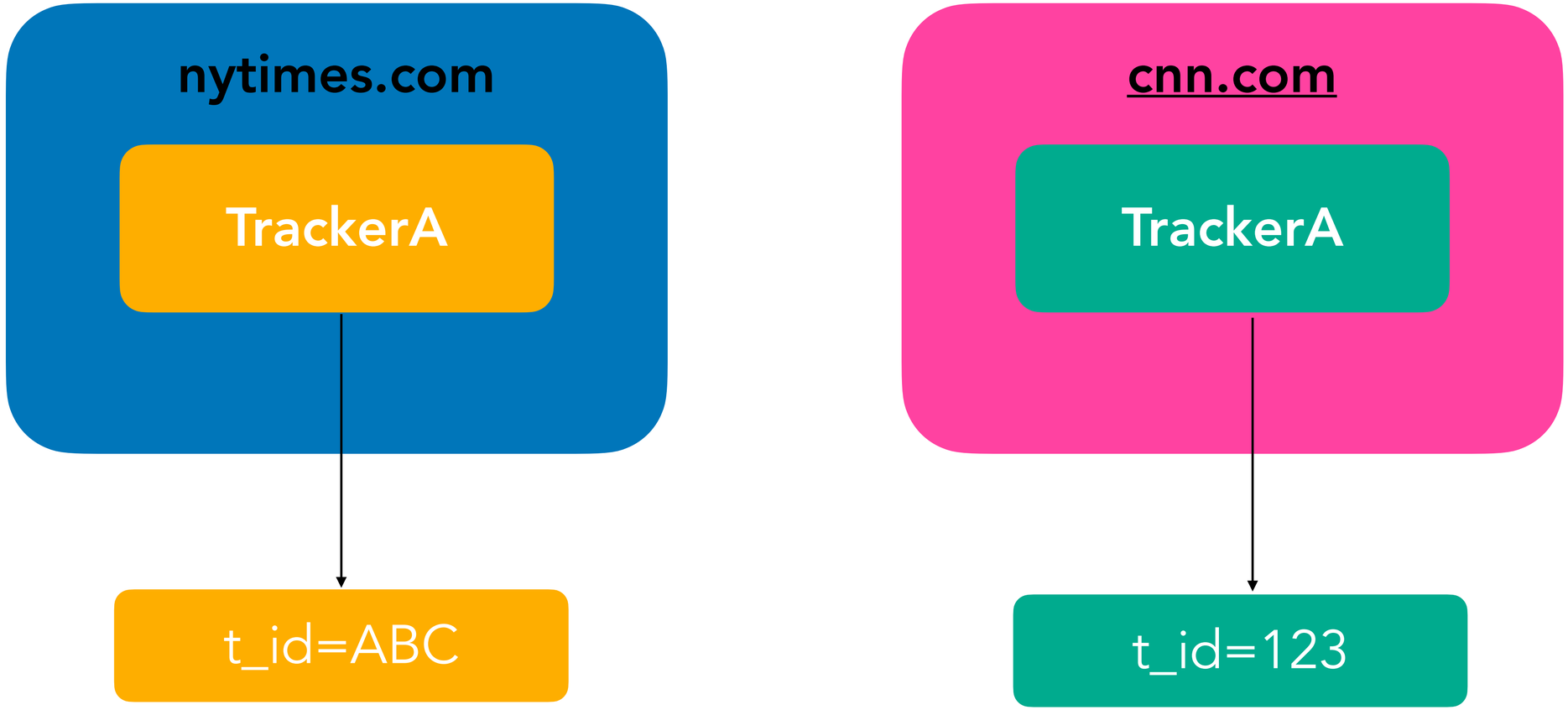
Third-Party Tracking



Third-Party Tracking



Partitioned Cookies



Why do Cookie Partitions Defeat Cookie Syncing?

- A tracker's *own* identifier is unique across different page visits, making cookie syncing *between* trackers essentially useless
- Cookie Partitioning is a **default** in most browsers!

Another way of beating cookie syncing… block third-parties

- The browser can automatically block third-party cookies from being sent!



Client

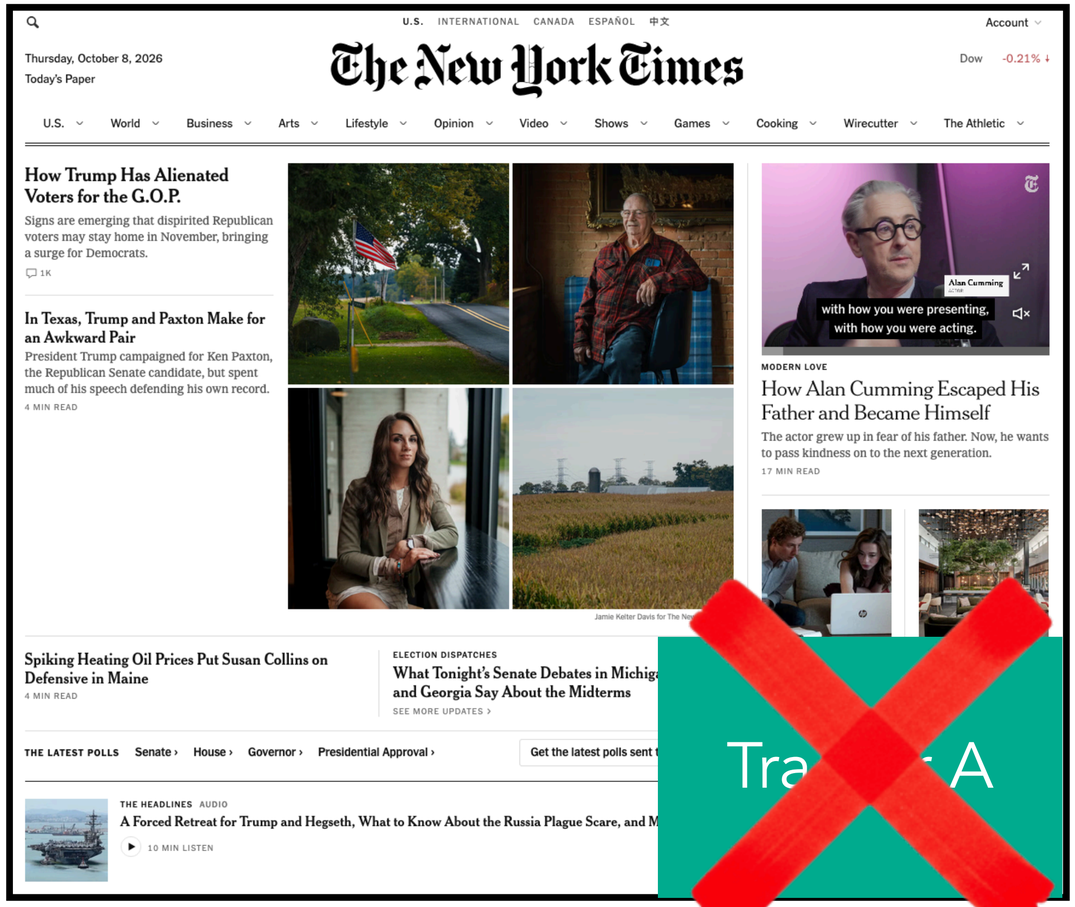
GET nytimes.com/

NYTimes Response

GET Track…A

Track… Pixel

Set-Cookie: t_id=ABC



Most browsers these days disable third-party cookies



Firefox



Brave



Safari

Most browsers these days disable third-party cookies



Firefox



Brave



Safari

Who is conspicuously missing from this list?

TECHNOLOGY

After 4 years, Google backtracks on its discontinuation of third-party cookies

JULY 27, 2024 · 7:51 AM ET

HEARD ON WEEKEND EDITION SATURDAY



Scott Simon

Google reneges on plan to remove third-party cookies in Chrome

By Aimee Picchi

Updated on: July 22, 2024 / 5:45 PM EDT / CBS News

Add CBS News on Google

ELECTRONIC FRONTIER FOUNDATION

About Issues Our Work Take Action Tools

Google Breaks Promise to Block Third-Party Cookies

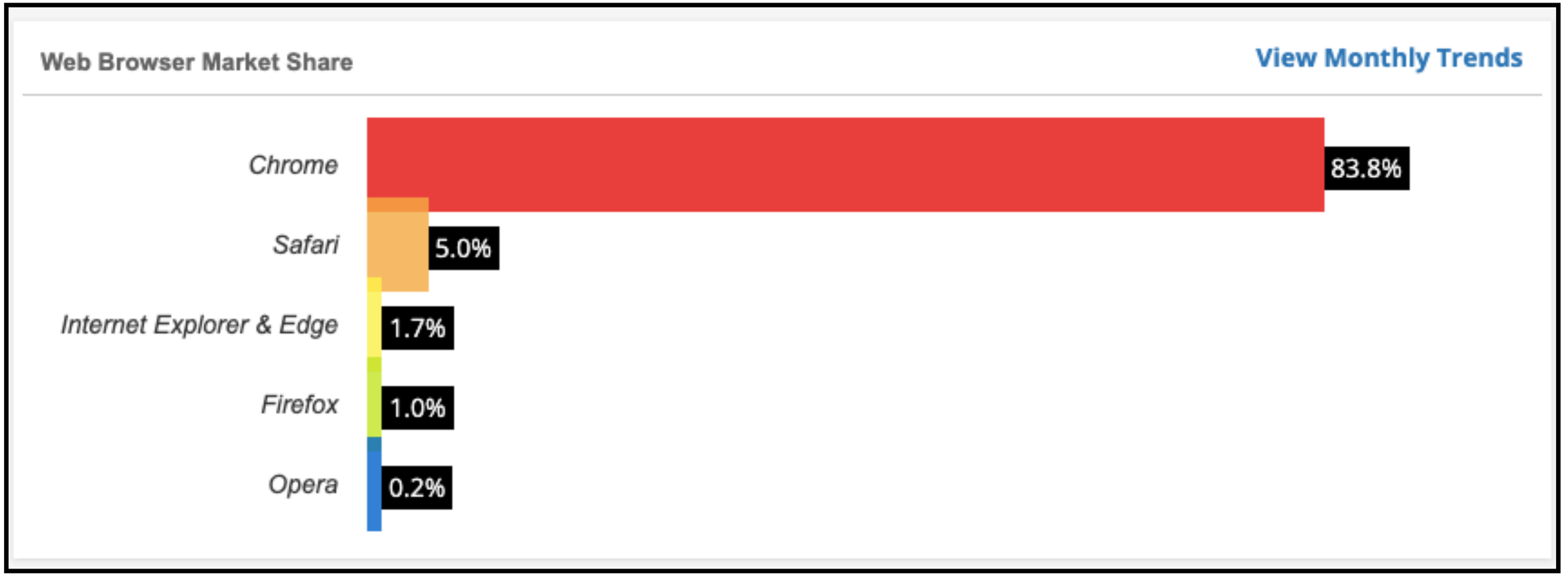
BY LENA COHEN | AUGUST 2, 2024

Chrome is so sad they can't block third-party cookies



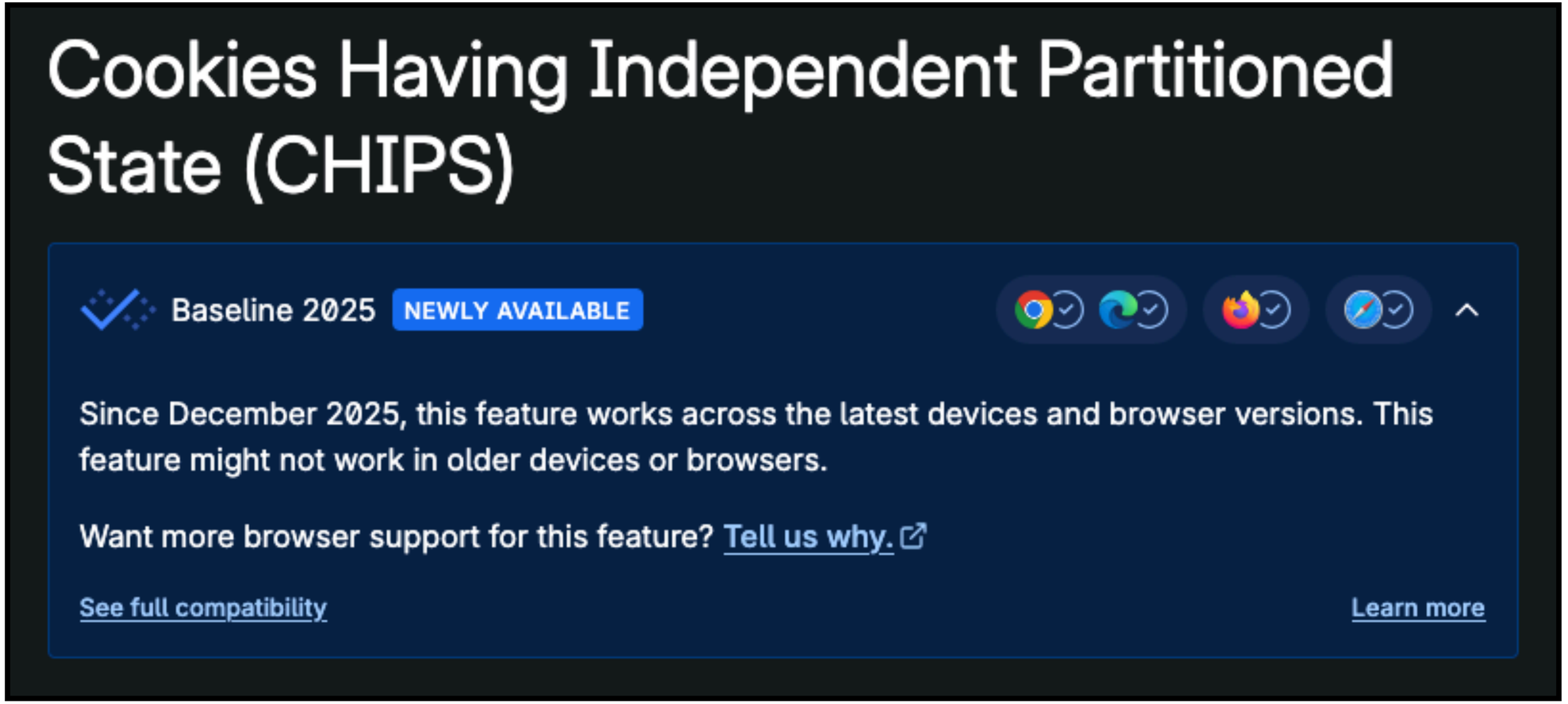
"At the same time, we recognize this transition requires significant work by many participants and will have an impact on publishers, advertisers, and everyone involved in online advertising."

Google Chrome is Keeping the Third-Party Tracking Dream Alive



Google implements partitioned cookies (if you turn them on)

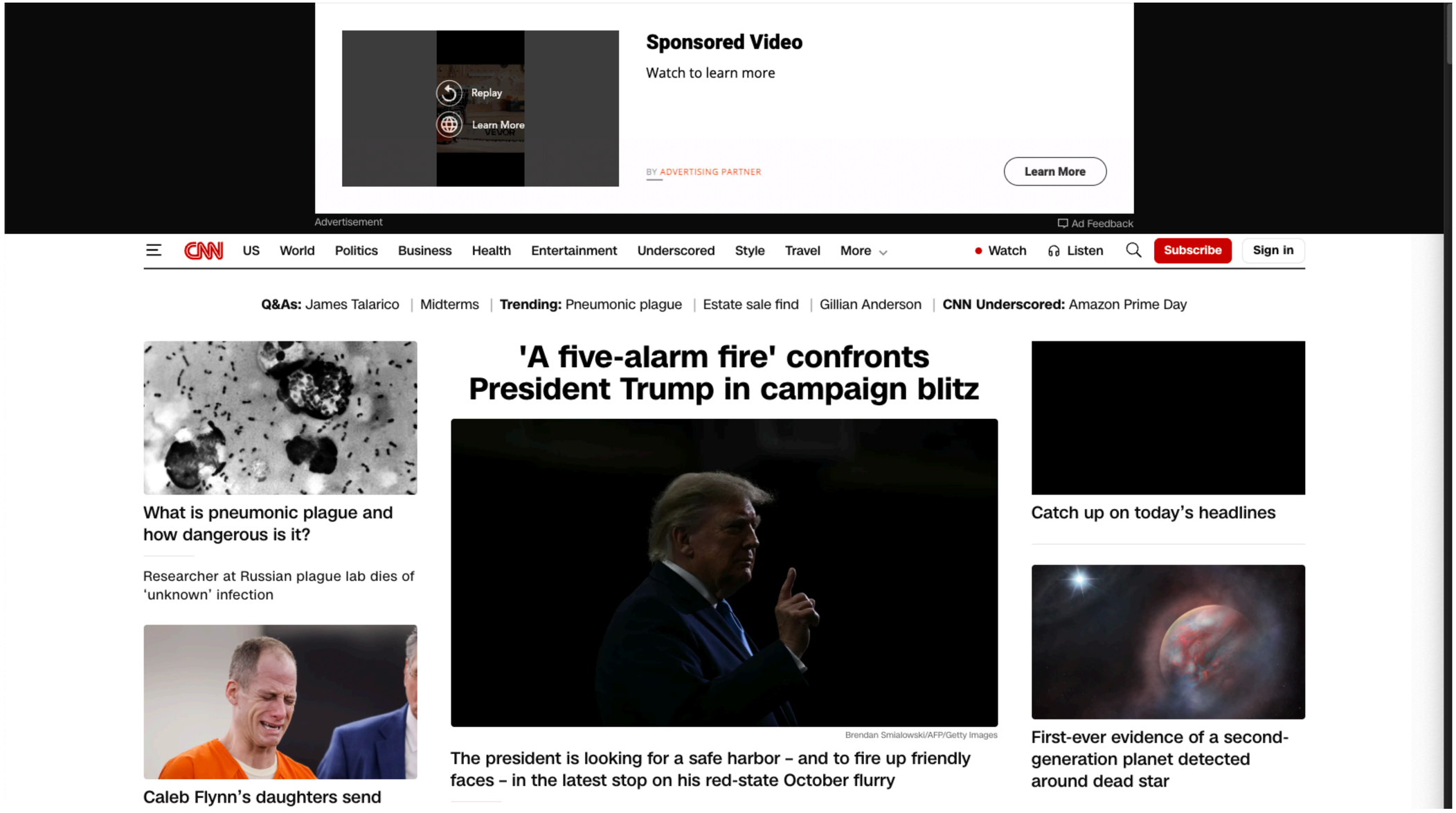
`Partitioned`



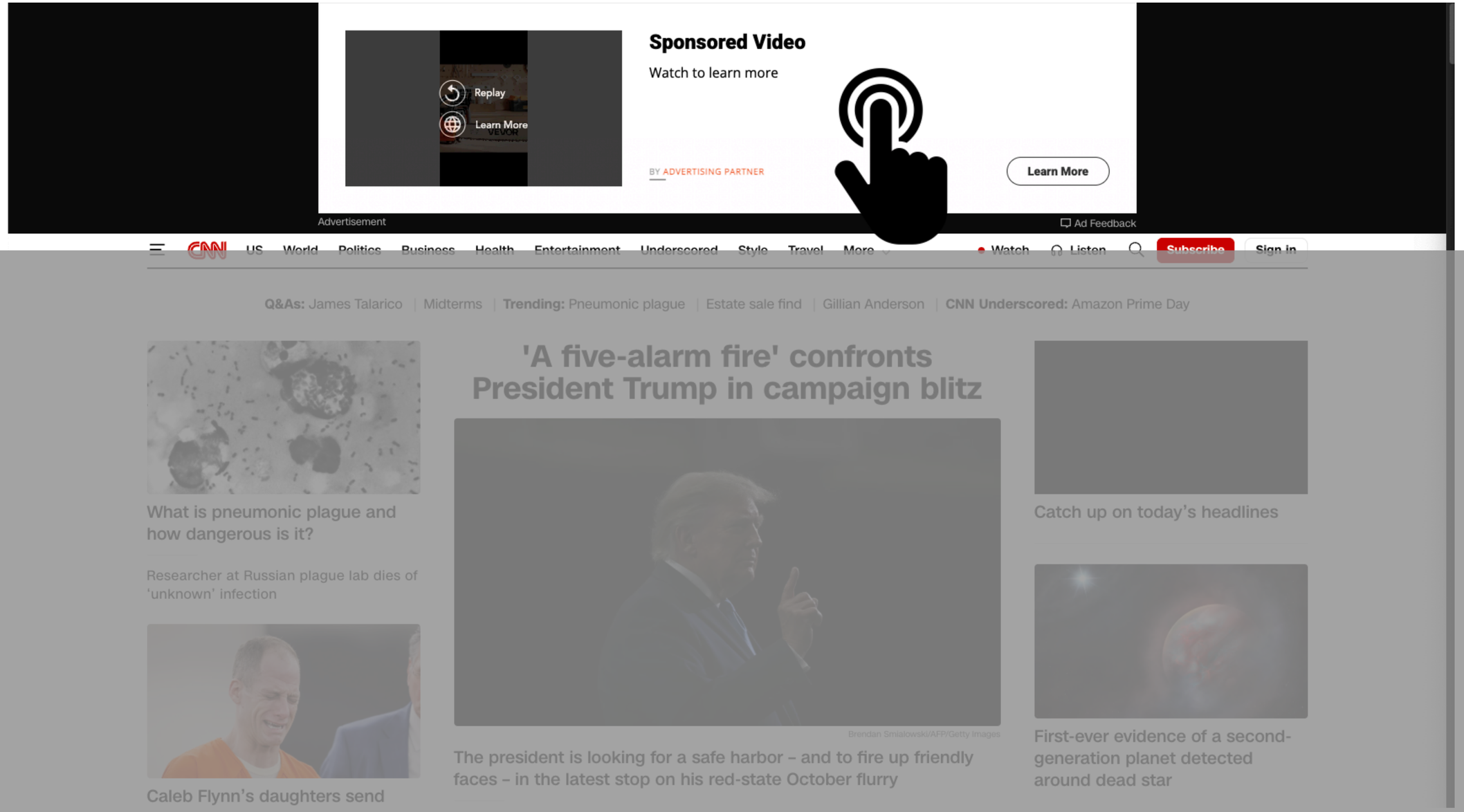
Trackers still need to contend with third-party cookies going away

- Why?
  - If you use a browser where they can’t track you
  - If you install an *extension* which stymies third-party blocking (e.g., most Ad blockers)
- So… what have trackers done about it?

Bounce Tracking



If I click on this link, where will it go?

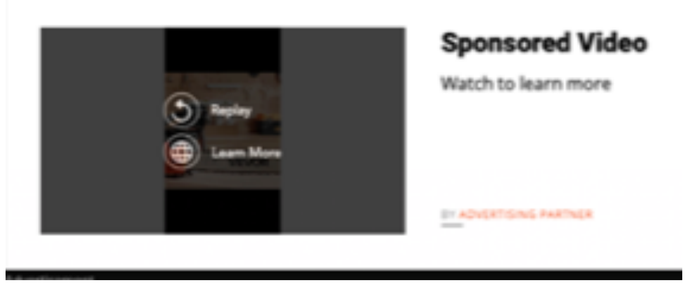


Bounce Tracking



Client

GET ad.com/?
redirect=tracker.com



tracker.com

ad.com

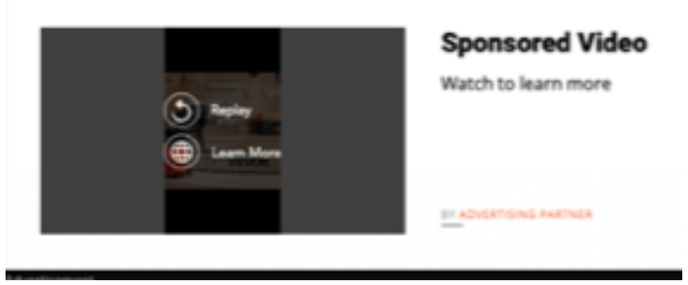
Bounce Tracking



Client

GET ad.com/?
redirect=tracker.com

302 tracker.com?
redirect=ad.com



tracker.com

ad.com

Bounce Tracking

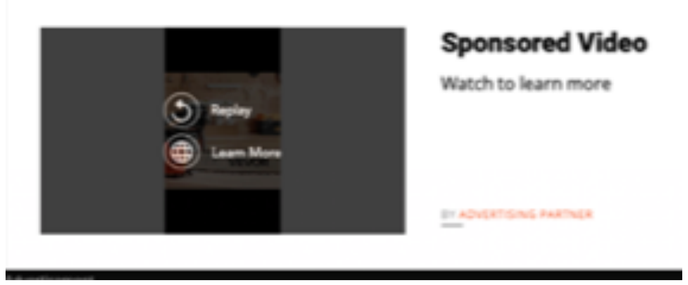


Client

GET ad.com/?
redirect=tracker.com

302 tracker.com?
redirect=ad.com

GET tracker.com/?
redirect=ad.com



tracker.com

ad.com

Bounce Tracking



Client

GET ad.com/?
redirect=tracker.com

302 tracker.com?
redirect=ad.com

GET tracker.com/?
redirect=ad.com

302 ad.com



tracker.com

ad.com

Bounce Tracking



Client

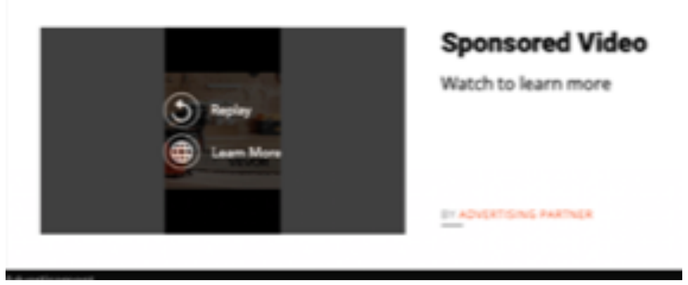
GET ad.com/?
redirect=tracker.com

302 tracker.com?
redirect=ad.com

GET tracker.com/?
redirect=ad.com

302 ad.com

GET ad.com



tracker.com

ad.com

Bounce Tracking



Bounce Tracking



Bounce Tracking



How does Bounce Tracking Bypass Third-Party Cookie Protections?

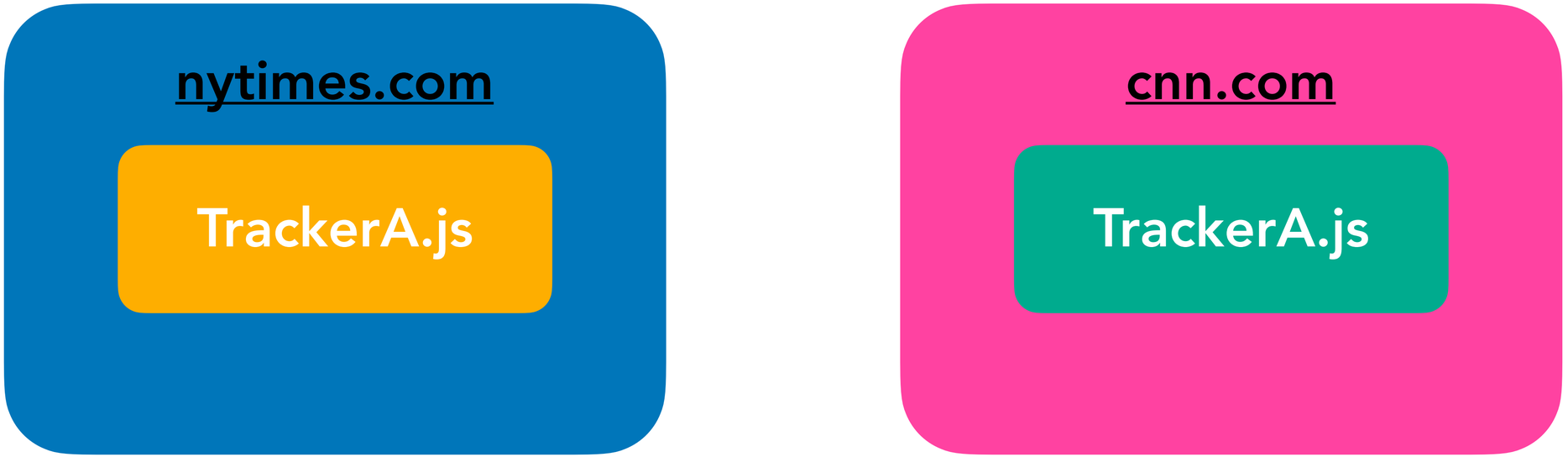
- Bounce Tracking *briefly* redirects your **entire page** to a different tracker, briefly giving them first-party access to your browser (and therefore, setting a cookie)
- Since browsers don't stop first-party requests, this remains totally possible today on most browsers
- Hard to defend against, since it's a "choice" — you could block redirects, but you might break the web
- But, OK, the trackers still have separate identifiers, right?

How does Bounce Tracking Bypass Third-Party Cookie Protections?

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- Hard to defend against, since it's a "choice" — you could block redirects, but you might break the web
- But, OK, the trackers still have separate identifiers, right?
  - **WRONG!**

UID Smuggling

- How can we smuggle identifiers from partitioned cookie stores?



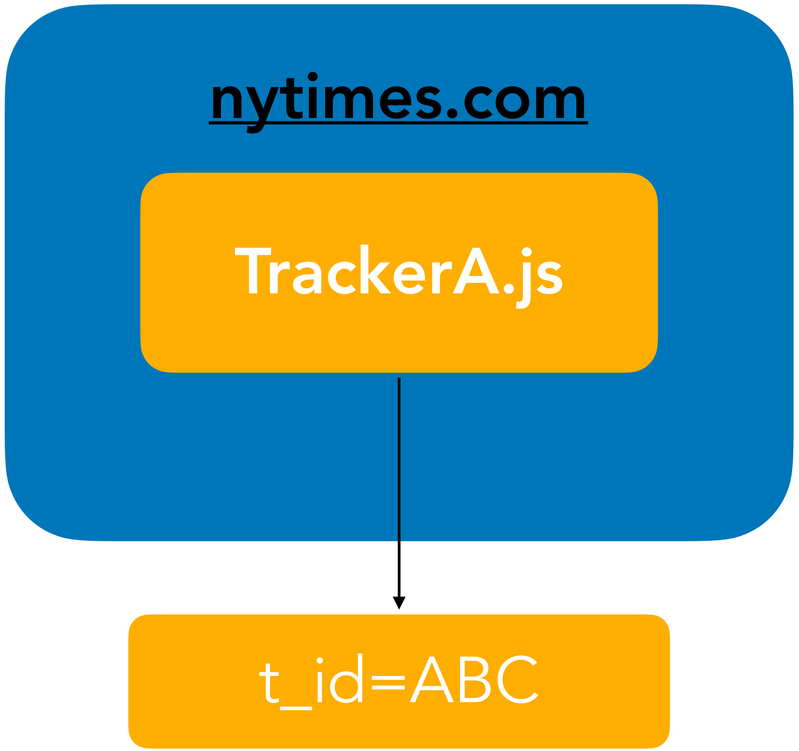
UID Smuggling

- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



UID Smuggling

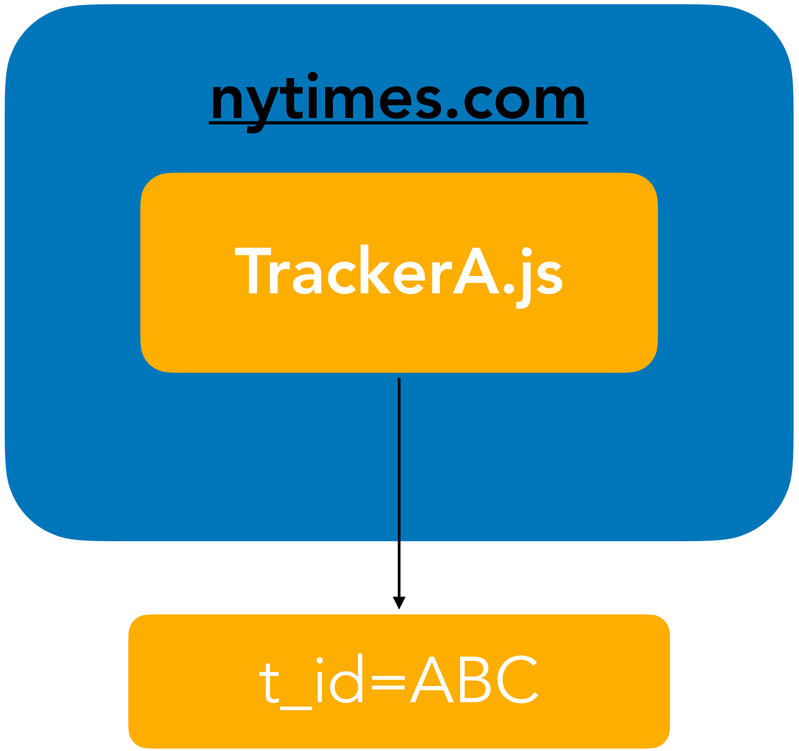
- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



What is the domain scope of this cookie set to?

UID Smuggling

- TrackerA.js is included on NYTimes page. This means TrackerA.js can set first-party cookies.



What is the domain scope of this cookie set to? nytimes.com + trackera.com

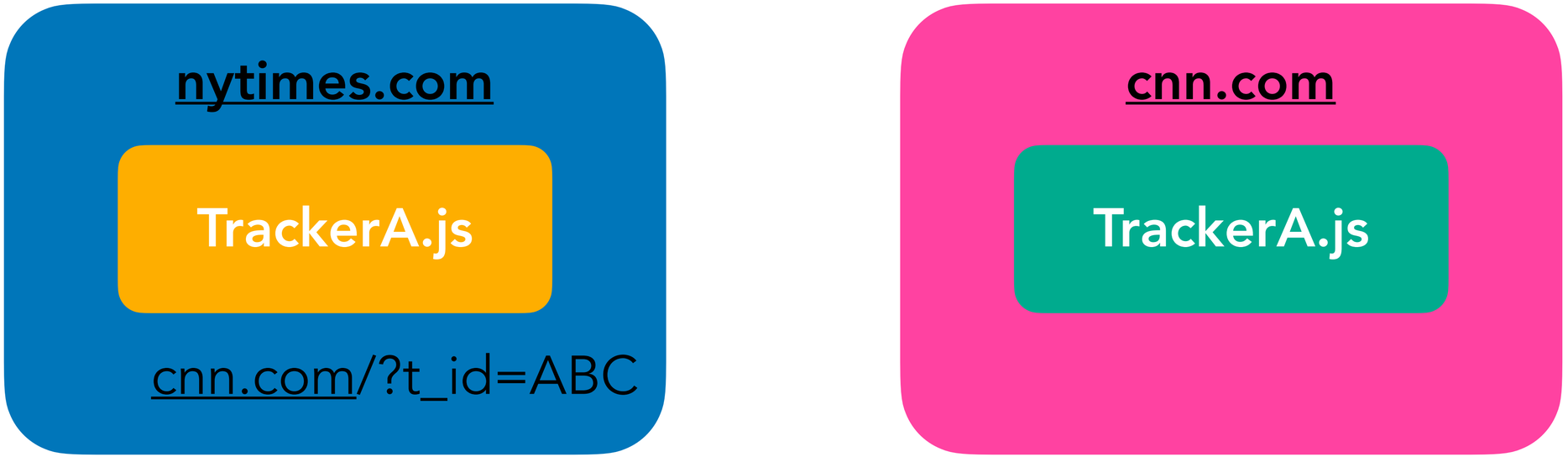
UID Smuggling

- Let's say there's a link to cnn.com on the nytimes.com, and TrackerA knows the JS also runs in cnn.com



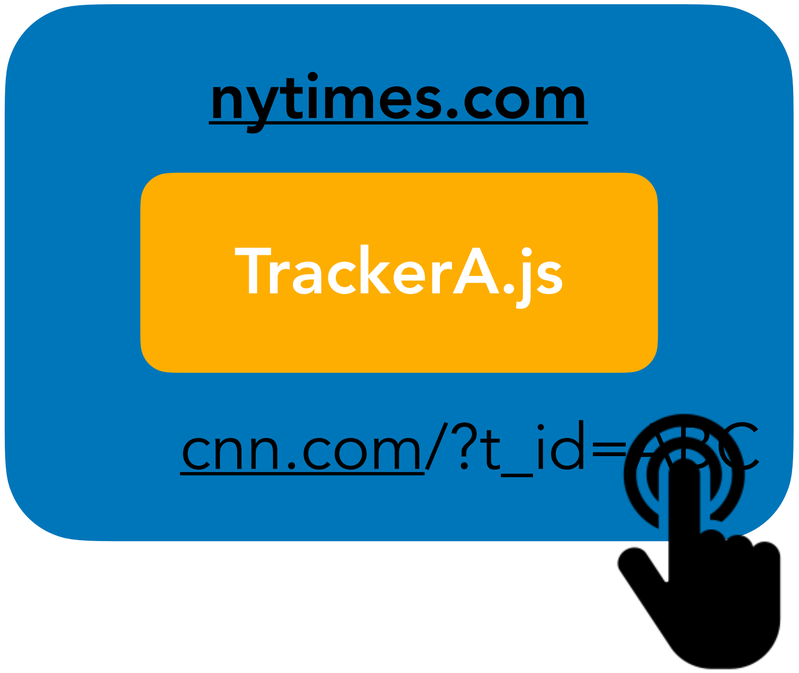
UID Smuggling

- TrackerA is going to rewrite the link to cnn.com to *smuggle* a UID.



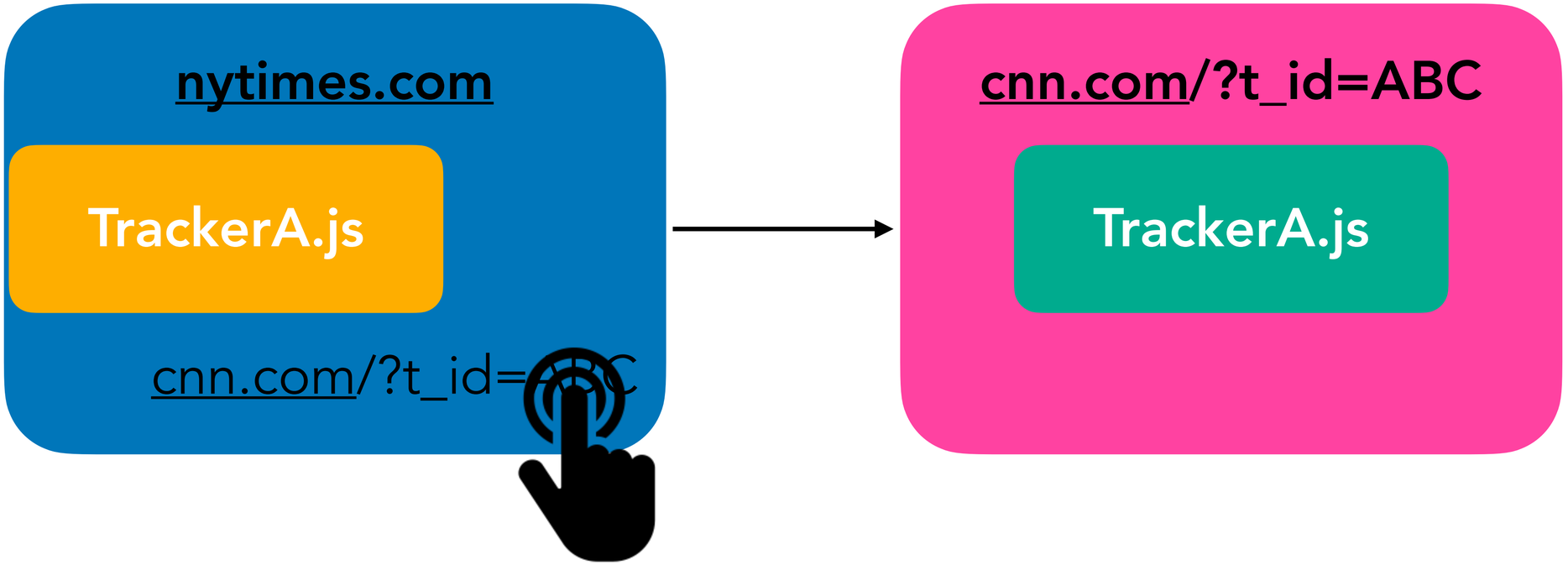
UID Smuggling

- When a user clicks on the link to cnn.com…



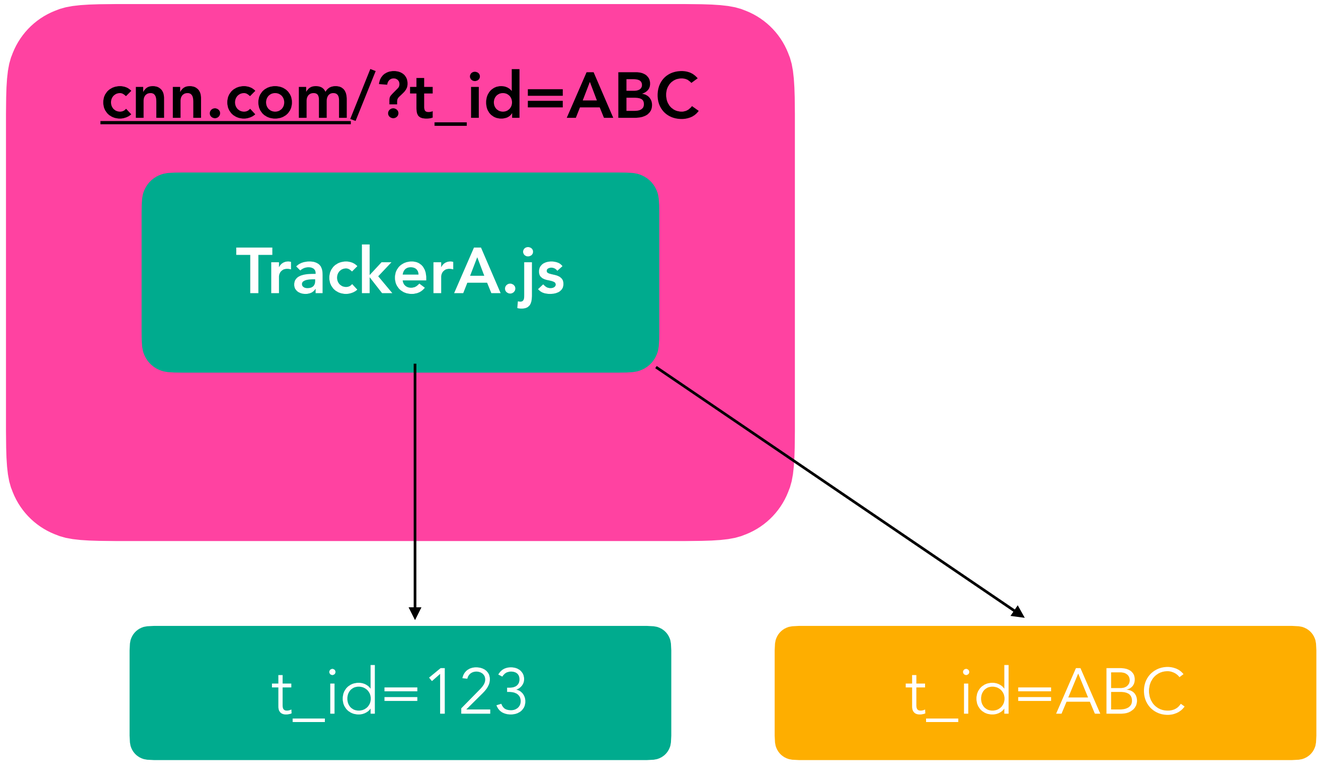
UID Smuggling

- When a user clicks on the link to cnn.com… they are routed directly to that page *with* the identifier in the URL



UID Smuggling

- TrackerA has its own cookie for cnn.com… but also, can read the UID from the URL bar! Now trackerA knows these identifiers are the same, and can link in a backend DB



Defeating UID smuggling

- Any ideas?

Defeating UID smuggling

- Any ideas?
- **Remove all query parameters from browser navigation.** Is this a good idea?
  - Might break the web. Might not. Could be interesting research!

Comparing Tracking Methods

	Cookie Syncing	**Bounce Tracking**	**UID Smuggling**
Browser mechanism	Tracker-to-Tracker redirect with syncURL	Routing users through redirects	Tracker appends ID to navigation URL
User Action?	Nothing!	User has to click on a link	User has to click on a link
Defense	Partition, Block Third-Party Cookies	Challenging to defend against	Remove all query parameters

Measuring UID Smuggling in the Wild

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Small aside: Pete from Brave will be giving a guest lecture on 11/3!

Other Strategies: CNAME Cloaking

- A CNAME is basically a pointer to another domain, tells the DNS hierarchy to “look over here” for the IP
- Basic idea — embed yourself **as a first-party** on the webpage by having a DNS record entry point to your domain
  - Bypasses third-party protection altogether

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```
11:16:51 kumarde@macbook ~ dig smetrics.cnn.com

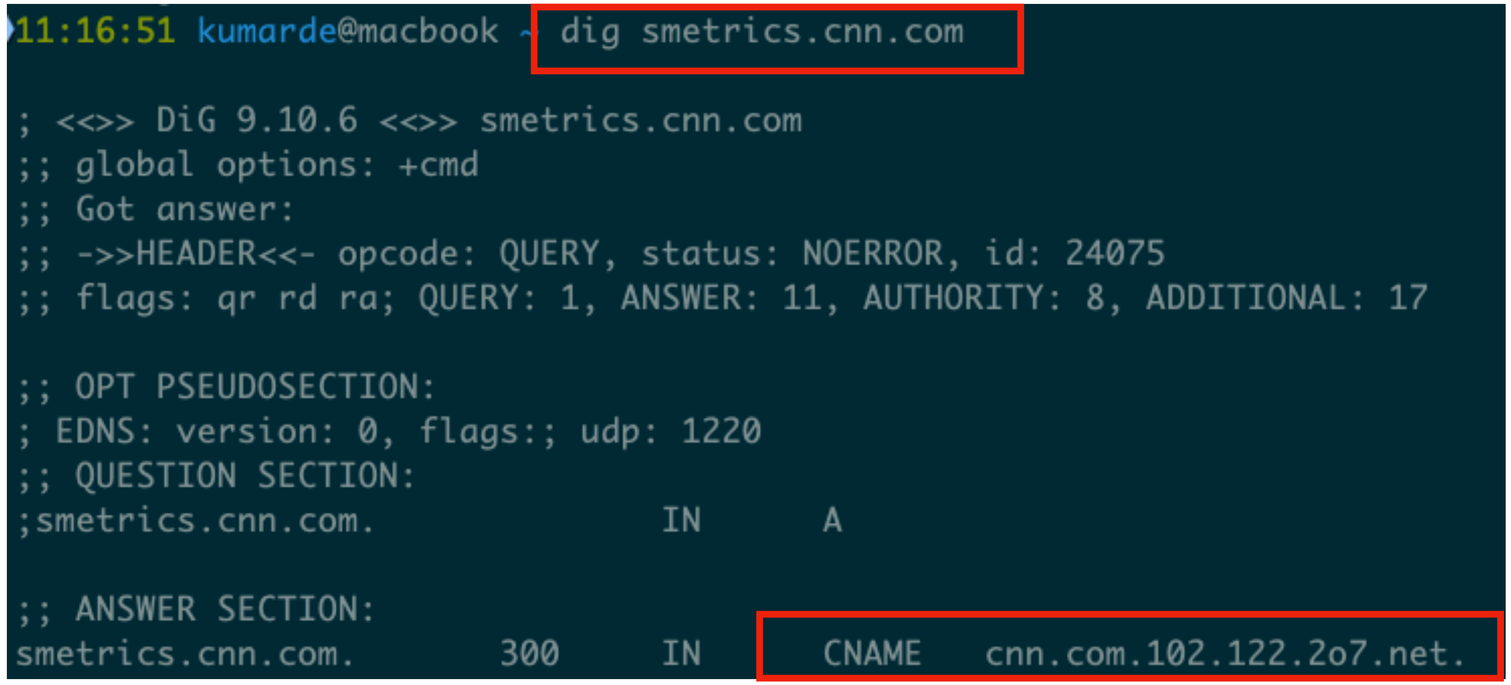
; <<>> DiG 9.10.6 <<>> smetrics.cnn.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24075
;; flags: qr rd ra; QUERY: 1, ANSWER: 11, AUTHORITY: 8, ADDITIONAL: 17

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;smetrics.cnn.com.              IN      A

;; ANSWER SECTION:
smetrics.cnn.com.       300     IN      CNAME   cnn.com.102.122.2o7.net.
```

Other Strategies: CNAME Cloaking

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; <<>> DiG 9.10.6 <<>> smetrics.cnn.com
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;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24075
;; flags: qr rd ra; QUERY: 1, ANSWER: 11, AUTHORITY: 8, ADDITIONAL: 17

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;smetrics.cnn.com.              IN      A

;; ANSWER SECTION:
smetrics.cnn.com.       300     IN      CNAME   cnn.com.102.122.2o7.net.
```

2o7.net is owned by Adobe Analytics… meaning CNN cookies are sent to Adobe by the browser by default :)

What can you do about it?

- Ultimately, your choice… but an easy way to control your privacy is to use a browser that values your privacy
  - Firefox, Safari, Brave, Tor (if you wish)… all of these do much better than Chrome by default
- Install ad blockers and privacy extensions!
  - UBlock Origin, Ghostery, Privacy Badger, Global Privacy Control… lots of ways to have a little bit more control over your data
- Stay informed!
  - Mozilla, Safari, Chrome, etc. all have blogs that regularly write about this stuff

Next time…

- **NO CLASS on 10/13!** Work on your PAs!
- Next Thursday, Kristen will be discussing inference… the basics of how companies use this data they collect about you to *actually infer things about you*!
- PA1 parts 3 + 4 will release next Wednesday!